

Brisa+ Technical Report

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Urban Morphology and CFD Patch Sampling for Wind Simulation Across Five Informal Settlements in Rio de Janeiro

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Document version	TR v1.0 (pre-CFD phase)
Pipeline version	v5.5 (May 2026 milestone — ROADMAP.md)
Build date	2026-05-03
Last numerical sweep	904040e (2026-05-03 — see .claude/agents/numerical-claims-auditor.md)
Author	Theo Hermann · MIT · thermann.ai@gmail.com
Repository	https://github.com/thrmnn/MorphoFavela
License	MIT (code) · CC-BY-4.0 (this report)
Audience	Engineering reviewers; researchers reproducing or extending the pipeline; CFD team consuming the patch contract. <i>Not</i> a journal manuscript — that lives separately and cites this report.

For the **fastest engineering review path**, see `docs/onboarding/engineering_review.md`. For terminology, see **§0 Glossary** below; for reproducibility, see **§12 Reproducibility**; for failure modes and observability, see **§13 Failure modes**.

Executive Summary

This report documents the morphometric analysis and CFD sampling pipeline developed for five informal settlements in Rio de Janeiro: Vidigal, Rocinha, Rio das Pedras, Complexo do Alemão, and Maré. The work supports the Brisa+ project, which aims to quantify pedestrian-level wind conditions in informal urban fabric and link them to health-relevant outcomes (thermal comfort, pollutant dispersion, natural ventilation).

For an engineering reviewer, in three sentences. *What to use:* the 119-patch CFD sampling campaign (per-patch `buildings.gpkg` + `terrain.tif` + `patch_meta.json`) and the morphometric grid (10 m, 20+ indicators per cell) are stable and ready to consume — every output path is documented in §8 and reproducible from §12. *What not to trust yet:* anything labelled "annualised" or "wind-velocity coupled" in §11, because no real OpenFOAM results have returned from the cluster — the result-side pipeline is shipped and synthetic-validated end-to-end on all 5 sites, but the producer interface has not been

exercised against real data. *What's pending:* VDG-P07 ingestion (in flight at MIT ORCD), followed by incremental submission of the remaining 118 patches and annualisation against the measured wind roses.

At time of writing, the pipeline has produced:

- **98,435 building footprints** compiled from site-specific and city-wide cadastral data, with defensive geometry repair and extended context beyond each favela boundary.
- **82,314 grid cells at 10 m resolution** covering the five sites, each characterised by 20+ morphometric indicators including Sky View Factor, plan area density, 8-direction frontal area density, volumetric porosity, height variability, slope, aspect, and street orientation entropy.
- **119 CFD simulation patches** selected via stratified sampling across twelve morphometric strata ($\text{SVF} \times \text{slope} \times \lambda_p$) with maximin geographic spacing and SVF-priority weighting to oversample health-relevant low-sky-view conditions.
- **A complete CFD integration pipeline** (`src/cfd_integration/`, 63 passing tests) ready to ingest OpenFOAM results when the simulation campaign completes in the parallel CFD repository. The result-side analysis pipeline (`scripts/analyze_cfd_results.py`) is shipped and synthetic-validated end-to-end on all 5 sites; awaiting the first real return (VDG-P07).

The pipeline is fully reproducible from the committed scripts; all inputs are documented and all intermediate outputs preserved in the canonical repository layout. This document summarises the methodology, validates the sampling design, and specifies the interface between this repository and the CFD execution environment.

0. Glossary / Nomenclature

Domain-specific terms used throughout this report. Each is defined once here, with units. Sections cite the term name; the definition lives only in this section.

Morphometric indicators (per 10 m grid cell)

Term	Symbol	Units	Definition
Sky View Factor	SVF	— (0–1)	Fraction of the upper hemisphere visible from a sample point at 1.5 m above ground. Cell value = mean over passageway samples whose centroid falls in the cell. Method: §4.2.
Plan area density (Building Coverage Ratio)	λ_p , BCR	— (0–1)	Sum of building footprint area in the cell ÷ cell area. Capped at 1.0 to prevent over-counting from overlapping footprints.
Frontal area density	λ_f	—	Projected vertical surface area per unit horizontal cell area, computed for 8 wind directions; stored as <code>lambda_f_{N,NE,...,NW}</code> , plus <code>lambda_f_mean</code> (mean across directions) and <code>lambda_f_max</code> (worst-case direction). Primary input to wind-canopy drag models.
Volumetric porosity	—	— (0–1)	$1 - (\Sigma \text{ building_volume_in_cell}) / (\text{cell_area} \times H_{\text{mean}})$. Strongly anti-correlated with λ_p ($r \approx -0.95$).
Height variability	σH	m	Standard deviation of building heights in the cell. NaN if < 2 buildings.
Mean building height	H_{mean}	m	Arithmetic mean of contributing building heights. Cell-level average — see §1 footnote on aggregation.
Max analysis-patch height	$H_{\text{max_analysis}}$	m	Tallest building inside a 100 m-diameter analysis patch (used for Blocken fetch check).
Slope, aspect	—	°, °	Cell-mean terrain slope and aspect from the merged DTM via numpy gradient. Aspect uses circular mean. Both are active predictors in the §5.3 aspect analysis (slope as scalar, aspect via sin/cos pair) and the §7.4 CFD regression (aspect adds an aspect → wind alignment covariate).
Street orientation entropy	—	— (0–1)	Normalised Shannon entropy of street-segment bearings (folded to [0, 180°)) within the cell. 0 = single direction, 1 = uniform.

Sky-view conventions

Term	Definition
Tregenza-145	Equal-area discretisation of the sky hemisphere into 145 patches (Tregenza, 1987). The IVF SVF engine ray-casts into the centroid of each patch from each sample point and reports the unobstructed fraction.
svfForProcessing153	UMEP's shadow-cast SVF processor with 153-patch sky discretisation (Lindberg & Holmer, 2010, used in SOLWEIG). Used as the independent benchmark in §4.2 / §10.3 cross-validation.
Height-matched	Both engines integrate at $z = 1.5$ m above ground, achieved in IVF via passageway sampling and in UMEP by lowering all building heights by 1.5 m before the shadow-cast.
Passageway aggregation	SVF cell value comes from samples on traversable ground (streets, alleys, courtyards) — never centroids inside building footprints, which would bias to 0.

CFD sampling

Term	Definition
Analysis patch	100 m-diameter circle (radius 50 m, area $\approx 7,854 \text{ m}^2$) where pedestrian-level metrics are sampled.
CFD domain	Per-direction rectangular <code>blockMesh</code> sized to $5/15/5/5 \cdot H_{\text{max}} + R_{\text{patch}}$ upstream/downstream/lateral/top, with a $5 \cdot W_{\text{patch}} = 500 \text{ m}$ lateral floor (Blocken 2015 wide-obstacle). One mesh per patch $\times$ wind-direction = 952 meshes for 119 patches $\times$ 8 directions. Canonical rule set: <code>src/cfd_integration/rectangular_domain_v1.json</code> .
12 strata	Cross-product of SVF (3 bins: < 0.15 , $0.15\text{--}0.30$, ≥ 0.30) $\times$ slope (2 bins: $< 15^\circ$, $\geq 15^\circ$) $\times \lambda p$ (2 bins: < 0.5 , ≥ 0.5). Stratum IDs encoded <code>SVFn_SLpN_LPn</code> .
Maximin spacing	Greedy maximin geographic distance between patch centres, with an 80 m floor. Produces a spatially diverse sample within each stratum.
Silhouette blockage	CFD blockage gating quantity. $B = D \cdot H_{\text{max}} / (2 \cdot \text{lateral} \cdot \text{top})$ — treats the 100 m analysis disk as a single solid block of height H_{max} (AIJ benchmark convention for wide-cluster CFD). Gate: $B < 0.05$ (Tominaga 2008). Uniform 2 % across the campaign once 5-W lateral floor dominates.
λ_F (frontal area density)	Per-direction $\Sigma(\text{projected_facade} \times \text{height}) / \text{disk_area}$, computed for 8 wind directions. The literature canopy-parameterisation form (Tominaga 2008 §3) — sums all facades, no shadowing, routinely > 1.0 in dense favela patches. Reported per-row alongside the blockage gate but <i>not</i> the gating quantity (silhouette envelope is).
Source-data envelope	The half-diagonal of the rotated worst-case rectangular domain plus a 50 m additive safety margin. Bounds the per-patch radius needed in <code>buildings_extended*.gpkg</code> and <code>dtm_extended*.tif</code> to cover all 8 wind-direction meshes. Range across the campaign: 565–646 m; <code>buildings_extended_700m.gpkg</code> covers all 119 patches.

Wind / atmosphere

Term	Definition
Wind rose	Per-site 8-direction frequency $\times$ mean-speed climatology, stored as <code>data/{site}/wind_rose.json</code> . Used post-hoc to weight the 8 directional CFD simulations into an annualised metric — <i>not</i> used for boundary conditions.
Calm fraction	Fraction of hourly observations with $ U < 0.5 \text{ m/s}$ or direction = NaN. Excluded from direction binning but recorded explicitly.
Neutral log-law	Inflow profile assumed for all CFD runs: $u(z) = (u_*/\kappa) \ln((z+z_0)/z_0)$. Implies neutral atmospheric stability; deliberate methodological simplification (§10.1).
k-ω SST	Shear-Stress-Transport k- ω turbulence closure (Menter, 1994). Standard for ABL-scale RANS.
ACH	Air change rate per hour: $\text{ACH} = 3600 \times \langle U \rangle / L$, with L the canyon characteristic length.
TI	Turbulence intensity: $\text{TI} = \sqrt{(2/3 \cdot \text{TKE})} / U_{\text{ref}}$.
TKE	Turbulent kinetic energy.
Stagnation fraction	Fraction of patch sample points with $ U < 0.5 \text{ m/s}$.

Data sources

Term	Definition
DTM	Digital terrain model (bare-earth raster).
DSM	Digital surface model (DTM plus buildings/vegetation).
INMET BDMEP	<i>Banco de Dados Meteorológicos para Ensino e Pesquisa</i> — the Brazilian Met. Service's hourly archive. Source for 4 of 5 wind-rose stations (A652, A636, A621, A602).
ASOS	Automated Surface Observing System — Iowa State's archive of international airport METAR; source for SBGL Galeão (Maré).
METAR	Standardised aviation weather report format.
EPSG: 31983	SIRGAS 2000 / UTM Zone 23S — the project's canonical projected CRS.

1. Study Sites

Five favelas were selected to span the morphological typologies of Rio informal settlements:

Site	Area	Type	Buildings (extended)	10 m cells	Mean building height †
Vidigal	0.30 km ²	hillside	4,600	3,169	6.3 m
Rocinha	0.80 km ²	hillside	14,443	8,972	8.1 m
Rio das Pedras	0.70 km ²	flatland	11,276	7,046	8.7 m
Complexo do Alemão	1.97 km ²	mixed	28,783	19,708	5.3 m
Maré	4.34 km ²	flatland	39,333	43,419	7.1 m
TOTAL	8.11 km²		98,435	82,314	—

Extended buildings = site footprints plus context buildings within a 300 m buffer (see Section 3.2). Typology assignment drives wind-regime analysis: hillside sites span 0–45° slopes; flatland sites cluster near 0°.

† Mean building height = mean of `H_mean` across all eligible 10 m grid cells per site (`outputs/{site}/morphometrics/grid/grid_metrics.gpkg`). This is a cell-area-weighted aggregation, not a per-building mean — short ancillary structures contribute less than the multi-storey blocks that fill more of the eligible cell area.

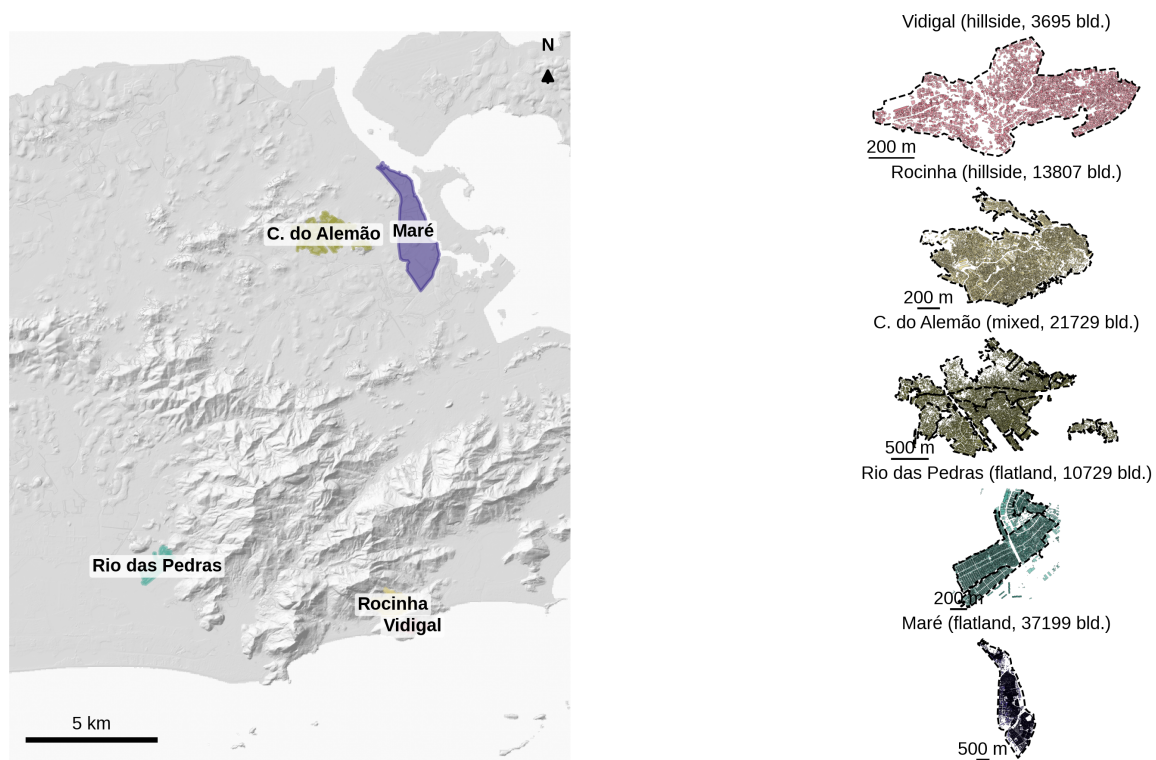


Figure 1. Overview of the five campaign sites on the Rio de Janeiro hillshade. The main panel shows the full metropolitan extent with all 1,074 registered favelas outlined in light grey; the five campaign sites are highlighted with typology-coloured fills. Right panels show each site's building footprints at a consistent visual style, with favela boundary and building count. Vidigal and Rocinha occupy the steep south-zone hillslopes near Sugarloaf; Maré and Complexo do Alemão are in the flat north-zone bayside plain; Rio das Pedras is in the west-zone plain.

Cidade de Deus was initially considered but excluded from the CFD campaign due to a documented building-data defect (see [.claude/.../memory/project_cdd_data_bug.md](#)); its SVF values saturated at ≈ 0.90 , indicating missing building footprints in the computational scene.

2. Data Sources

2.1 Site-specific data

Each site has a `data/{site}/raw/` directory containing:

- **Building footprints** (.shp, 3D polygons) with attributes `altura` (height above base), `base` (base elevation), `topo` (top elevation). CRS: EPSG: 31983 (SIRGAS 2000 / UTM zone 23S).
- **Favela boundary** (.shp) from the 2019 Rio municipal cadastre.

- **Digital terrain model** (.tif) at 5 m resolution clipped to the site extent during data preparation (manually in GIS; not regenerated code-side per project memory).
- **Road network** (.shp) from the municipal Logradouros dataset, clipped and projected per site.

2.2 City-wide data (shared across sites)

Used to extend the building context and ensure continuous terrain for CFD domains that exceed individual favela boundaries.

- **data/RJ/buildings_RJ_2019.shp**: 2,362,806 building footprints for the entire municipality. Same schema as per-site files (subset extracted from this source).
- **data/RJ/DTM_RJ.tif**: 72 km × 37 km DTM at 5 m resolution covering all of Rio.
- **data/RJ/Favelas_Limit_2019.shp**: 1,074 registered favelas, used for the Figure 1 overview.

2.3 Wind forcing (INMET)

Annualised wind roses are required to weight CFD results across the eight cardinal directions. Each site has a `data/{site}/wind_rose.json` file whose schema is defined by `src.cfd_integration.schema.WindRose` — frequencies and mean speeds per direction, plus provenance metadata (station id, coordinates, time window, observation count, calm fraction, anemometer reference height, and a `quality_flag` ∈ {`measured`, `gap-filled`, `placeholder-prior`}).

Current status (April 2026). All five `wind_rose.json` files now carry measured hourly observations across the full 2015–2024 window and are tagged `quality_flag: "measured"`. The four hillside / inland sites use INMET BDMEP records; Maré uses the Iowa State ASOS METAR archive of SBGL Galeão. Sample sizes range from $n = 64,088$ (A636 Jacarepaguá, which only began reporting in August 2017) to $n = 89,439$ (SBGL). Per-site directional frequencies, mean speeds, and calm fractions are shown in **Figure S5** (`docs/technical_report/figures/figS5_wind_roses.png`):

Site (station)	n	Window	Calm	Prevailing dir	Notes
Vidigal / Rocinha (A652 Forte de Copacabana)	85,103	2015–2024	1.4 %	E (30 %), W (23 %)	Coastal cliff: bimodal sea-breeze and reverse-channel along the Dois Irmãos saddle.
Rio das Pedras (A636 Jacarepaguá)	64,088	2017–2024	46.3 %	SW (26 %), W (20 %)	Sheltered basin, weak winds ($\bar{U} \approx 1.3$ m/s); high calm fraction reflects the Jacarepaguá lowland regime, not measurement gaps.
Complexo do Alemão (A621 Vila Militar)	86,019	2015–2024	33.1 %	N (25 %), SW (19 %)	North-zone interior; bay sea-breeze + post-frontal SW pattern.
Maré (SBGL Galeão METAR)	89,439	2015–2024	3.7 %	SE (26 %), E (19 %)	Bay regime; continuous METAR coverage gives the largest sample.

Recommended stations (verified April 2026 against the INMET catalogue, daily-graph URLs, and the published TMY paper for A652):

Site	Station	Code	Coords (lat, lon)	Class	Notes
Vidigal	Forte de Copacabana	A652	–22.988, –43.190	coastal	Nearest unobstructed coastal reference (~5 km E). CFD captures the Dois Irmãos lee-side locally; inflow rose stays at A652.
Rocinha	Forte de Copacabana	A652	–22.988, –43.190	coastal	Provides the unobstructed SE → NE driver. Valley channelling is resolved by the CFD itself.
Rio das Pedras	Jacarepaguá	A636	–22.99, –43.37	plain	Colocated with the Jacarepaguá lowland.
Complexo do Alemão	Vila Militar	A621	–22.86, –43.41	urban interior	Closest north-zone station (~8 km W). Corrects the earlier placeholder recommendation of A602 Marambaia, which is geographically mismatched (southwest coast, not north zone).
Maré	SBGL Galeão METAR (preferred); A652 (INMET fallback)	— / A652	–22.81, –43.25 / –22.988, –43.190	bayside	Galeão airport METAR via the Iowa State ASOS archive is the best match for the bay regime; ingested via <code>build_wind_rose.from_iowa_asos_csv</code> . A652 is the INMET fallback.

The earlier placeholder attributed A652 to "Alto da Boa Vista" — that name belongs to the municipal Alerta Rio network, not INMET. A652 is Forte de Copacabana. The station table is corrected here.

Data acquisition. INMET publishes one nationwide yearly ZIP (~100 MB) at `https://portal.inmet.gov.br/uploads/dadoshistoricos/{YEAR}.zip`, live and unauthenticated (browser User-Agent required). Each ZIP contains one CSV per automatic station for the year. The ingestion pipeline is documented in `src/cfd_integration/README.md` and implemented in `scripts/build_wind_rose.py` from `inmet_csv`.

BDMEP CSV format: `sep=;`, `decimal=,`, latin-1 encoding, 8-row metadata header, missing values encoded as `-9999`, anemometer at $z = 10$ m. Calm periods ($|U| < 0.5$ m/s or `direction = NaN`) are excluded from direction binning but recorded in `calm_fraction` so they are not hidden.

Neutral-stability assumption. The k - ω SST inlet uses a log-law profile that implicitly assumes a neutral atmospheric boundary layer. For Rio, daytime unstable convection and evening stable inversions bias the stagnation metric. This limitation is accepted for the screening campaign and documented in §10. A future campaign could stratify the rose by stability class (e.g., day vs night, or Pasquill-Gifford class).

2.4 Coordinate reference system

All vector and raster data are in **EPSG:31983 (SIRGAS 2000 / UTM zone 23S)**, verified on load. Mixed-CRS inputs are reprojected automatically in the pipeline.

3. Data Preparation

3.1 Building footprint processing

Raw footprints undergo three processing steps before use:

1. **Topology validation.** Shapely's `is_valid` flag is checked on all geometries. Rocinha was found to have 10 self-intersecting polygons (0.07 %) out of 13,807; all other sites had zero invalid geometries. All invalids are self-intersections (common artefact of municipal GIS digitisation) and are repaired by `buffer(0)`, which corrects the geometry with sub-square-metre area change. The repair is applied defensively in both the extended-context builder and the morphometric pipeline before any spatial operation. Full audit: `outputs/comparative/audits/rocinha_topology_audit.md`.
2. **Height validation.** `altura` values are checked for negatives (none) and unrealistic extremes (one building > 50 m — a legitimate high-rise in context data near a favela).
3. **CRS enforcement.** If source CRS differs from EPSG:31983, the dataset is reprojected.

3.2 Extended building context

Problem. Each CFD simulation needs building + terrain context out to the rectangular per-direction domain envelope (§6.5, `rectangular_domain_v1.json`). The rotated worst-case rectangle has half-

diagonal 565–646 m across our 119-patch campaign — the campaign-maximum is RDP-P15 at 646 m ($H_{max_analysis} = 27.3$ m). A naïve sampling restricted to patches fully within the favela excludes 55 % of candidate cells at Vidigal.

Solution. Two passes:

1. **Initial 300 m extension** (used by the original eligibility filter and SVF cross-validation). The city-wide RJ building dataset is clipped to a 300 m buffer around each favela boundary and merged with the site-specific dataset, preferring site data where footprints overlap (deduplication via spatial join). This is the buffer documented in Figure S2.
2. **700 m extension for v1 CFD domains** (May 2026 migration). The same merge, larger buffer — covers the rectangular source-data envelope of every campaign patch, including RDP-P15 at 646 m. All five sites have both `buildings_extended_300m.gpkg` and `buildings_extended_700m.gpkg` (and matching DTMs); the CFD-side `build_patch_case.py` uses the 700 m product.

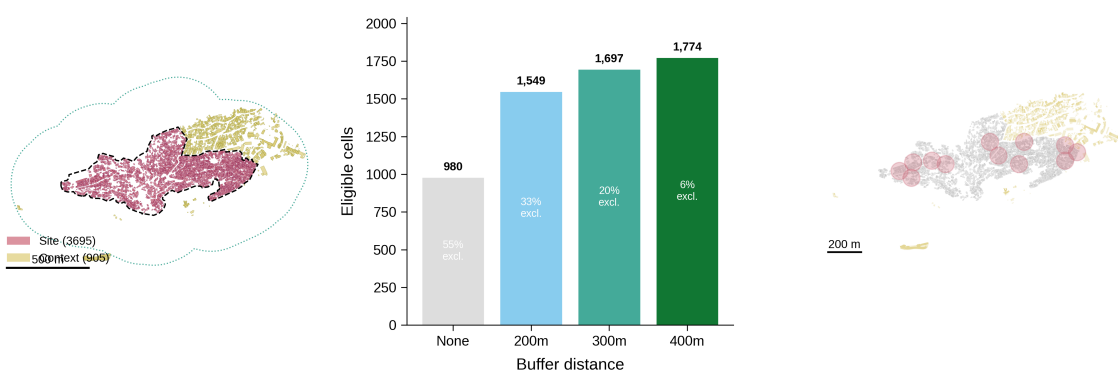


Figure S2. Context extension for Vidigal at the original 300 m buffer. Left: site buildings (blue) within the boundary and context buildings (orange) from city-wide data within the 300 m buffer. Middle: candidate-pool sensitivity to buffer distance — 0 m yields 980 eligible cells (31 %); 300 m yields 1,697 (54 %); 400 m only adds 77 more (diminishing returns at the eligibility-filter level). Right: pilot patch locations on the extended footprint — without the extension, patches cluster on the interior; with it, they span the full favela plus its immediate fringe. The 700 m v1 buffer extends the same construction further; counts grow to 10,713 buildings at Vidigal, 46,390 at Maré.

Extended buildings per site:

Site	Site buildings	Context buildings	Extended total
Vidigal	3,695	905	4,600
Rocinha	13,807	636	14,443
Rio das Pedras	10,729	547	11,276
Complexo do Alemão	21,729	7,054	28,783
Maré	37,199	2,134	39,333

Implementation: `scripts/build_extended_context.py`. The 300 m buffer was selected as the inflection point of a sensitivity scan (200 m / 300 m / 400 m); 300 m reduces domain-extent exclusion to $\leq 20\%$ at all sites while adding modest file size.

3.3 Extended terrain

The site-specific DTM is merged with the city-wide RJ DTM (both 5 m resolution) over the same 300 m buffer. The site DTM takes priority in overlap regions (`rasterio.merge.merge` with `method="first"`). A post-merge sanitisation step replaces any residual float32-max sentinel values (historical nodata artefact) with a clean nodata = -9999 and declares it in the output profile. This was found to be necessary after the first run produced downstream warnings about values of order 3.4×10^{38} in the merged raster.

4. Morphometric Grid

4.1 Grid construction

A regular 10 m grid is generated covering each extended-buildings extent, then clipped to the favela boundary so that only cells whose centroid falls within the settlement are retained. Cells range from 3,169 at Vidigal to 43,419 at Maré, with total 82,314 cells across the five sites.

4.2 Indicators

Per 10 m cell, the pipeline computes:

Sky View Factor (SVF). The fraction of hemispheric sky visible from pedestrian height. Computed at 1.5 m above the ground surface via Tregenza 145-patch ray-casting on a sample grid of passageway points; the per-cell value is the mean of all sample points whose centroid falls within the cell. An `svf_count` column records the number of contributing sample points per cell (mean per site ranges from 1.9 (Maré) to 12.9 (Vidigal); pooled mean 3.5; minimum 1). Cells with no contributing samples receive NaN. The cross-site spread reflects

passageway density: hillside fabrics with narrow, frequent corridors yield more samples per cell than flatland blocks where samples concentrate in a few wide streets.

Critically, SVF is **aggregated from passageway samples, not computed at grid-cell centroids**. Centroid computation would bias SVF toward 0 for cells whose centroid falls inside a building; passageway aggregation avoids this artefact. A diagnostic verification confirmed no spurious SVF = 0 spikes (see `src/morphometry/grid.py::_aggregate_svf_to_grid` and `docs/technical_report/validation.md`).

Cross-validation against UMEP. The Tregenza-145 ray-cast SVF was benchmarked against UMEP's shadow-casting SVF processor (`svfForProcessing153`, Lindberg & Holmer 2010, used in SOLWEIG) on a 1 m digital surface model rasterised from the same building footprints and DTM. UMEP averages over 153 hemispherical patches via shadow-casting on the DSM rather than ray-casting on a 3D mesh; this is an algorithmically distinct independent reference. To make the two engines height-comparable, we lower every building height by 1.5 m before running UMEP — equivalent to lifting the integration plane to pedestrian height — and mask rooftop pixels (UMEP otherwise includes them, IVF does not because passageway samples never fall on roofs). Aggregating UMEP at the same 10 m grid across all five sites:

Site	n	r ²	slope	RMSE	bias (UMEP – IVF)
Vidigal	2,510	0.68	0.96	0.12	+0.01
Rocinha	5,646	0.81	1.01	0.14	+0.09
Rio das Pedras	1,503	0.93	1.25	0.11	+0.08
Complexo do Alemão	4,838	0.76	0.92	0.17	+0.14
Maré	9,516	0.94	0.97	0.12	+0.09

Across all five sites the engines agree on the same physical quantity: four of five regression slopes (Vidigal 0.96, Rocinha 1.01, Complexo do Alemão 0.92, Maré 0.97) fall within $\pm 8\%$ of unity. Rio das Pedras is the slope outlier at 1.25 — the site has the smallest valid grid ($n = 1,503$ after eligibility filtering), and its eligible cells are concentrated in the 0.3–0.5 SVF range where the 153-patch shadow-casting integration picks up partial obstruction that the 145-patch ray-cast smooths through. This is a sensitivity to grid coverage at smaller settlement extents, not a discrepancy in how the two engines compute SVF.

The r^2 range (0.68–0.94) tracks site relief: low-relief, spatially coherent fabrics (Maré 0.94, Rio das Pedras 0.93) yield the strongest per-cell agreement because adjacent cells share local terrain and similar average obstruction. High-

relief hillside settlements (Vidigal 0.68, Complexo do Alemão 0.76, Rocinha 0.81) produce more cell-to-cell variation between integrators because each cell sits in a distinct local viewshed.

All five biases are positive (UMEP integrates over more sky than IVF), ranging from +0.01 (Vidigal) to +0.14 (Complexo do Alemão). The direction is consistent with the +1.5 m raised-observer transform applied to building heights before running UMEP: structures shorter than 1.5 m get partially or fully zeroed in IVF's input, opening additional sky in the UMEP comparison. Sites with substantial single-storey residential coverage (Complexo do Alemão, Maré, Rocinha, Rio das Pedras) sit in the +0.08 to +0.14 range; Vidigal's +0.01 reflects a footprint dominated by taller multi-storey blocks where the height shift has limited effect. The Vidigal bias collapse from -0.05 ($z = 0$, prior comparison) to $+0.01$ ($z = 1.5$) confirms the systematic offset is the sampling-height difference rather than the algorithmic distinction between ray-casting on a 3D mesh and shadow-casting on a DSM. See `scripts/validate_svf_against_umep.py` and `outputs/{site}/morphometrics/svf/umep_validation/`.

Plan area density (λ_p , BCR). Building footprint area $\div$ cell area. Capped at 1.0 to prevent over-counting from overlapping footprints in dense informal fabric.

Frontal area density (λ_f). Projected vertical surface area per unit horizontal cell area, computed for eight wind directions (N, NE, E, SE, S, SW, W, NW); stored as `lambda_f_{direction}` plus `lambda_f_mean` and `lambda_f_max`. Directional λ_f is the primary morphological input to wind-canopy drag models.

Volumetric porosity. $1 - (\sum \text{building_volume_in_cell}) / (\text{cell_area} \times H_{\text{mean}})$. Characterises the void fraction in the urban canopy; strongly anti-correlated with λ_p ($r \approx -0.95$ across sites, see Section 5.3).

Height variability (σ_H). Standard deviation of building heights in the cell. Cells with < 2 buildings receive NaN.

Mean height (H_{mean}). Arithmetic mean of contributing building heights.

Street orientation entropy. Shannon entropy of street-segment bearings clipped to the cell (bearings folded to $[0, 180^\circ)$ because streets are bidirectional), normalised to $[0, 1]$. Measures how ordered the street grid is: 0 = single dominant direction, 1 = uniform distribution across 36 bins.

Terrain slope and aspect. Derived from the merged DTM via numpy gradient, sampled within each cell and averaged. Aspect uses a circular mean to avoid discontinuity at $0^\circ/360^\circ$. Aspect is a circular quantity and is encoded as the orthogonal pair $(\sin \alpha, \cos \alpha)$ whenever it enters a linear regression, so that a north-facing cell at $\alpha = 359^\circ$ and one at $\alpha = 1^\circ$ fall close in the predictor space

rather than at opposite ends. The same module also exposes a quadrant binner (N / E / S / W centred on cardinals, edges at the inter-cardinals) for stratified summaries (`src/morphometry/aspect.py`).

Implementation: `src/morphometry/grid.py`. The computation is deterministic and takes ~30–80 s per site (longest is Complexo do Alemão at ~85 s) on standard hardware.

4.3 Distributions across sites

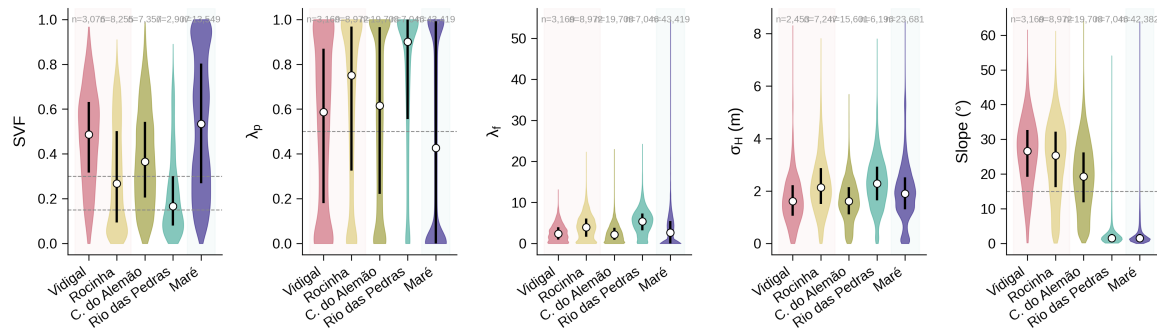


Figure 2. Distributions of the five primary indicators (SVF, λ_p , λ_f , σ_H , slope) across all five sites. Violins show the full distribution per site; white dots are medians and dark bars span the IQR. Dashed horizontal lines mark stratum boundaries (SVF at 0.15 and 0.30; λ_p at 0.5; slope at 15°). Background shading groups hillside (left, warm) and flatland (right, cool) sites.

Key patterns:

- **Slope clearly separates typologies.** Hillside sites (Vidigal, Rocinha) have most cells at 15–35°; flatland sites (Rio das Pedras, Maré) concentrate below 5°; Complexo do Alemão spans both regimes.
- **λ_p medians span 0.43 (Maré) to 0.90 (Rio das Pedras)** — denser than is typical of formal urban fabric. Maré, with its planned-housing block interiors, sits at the low end; Rio das Pedras, the densest informal infill, at the high end.
- **SVF distributions differ substantially.** Rocinha's dense canopy produces a strongly left-skewed distribution (median 0.27); Maré's more open layout has a right-skewed distribution (median 0.53).
- **σ_H is small everywhere** (median 1.6–2.3 m, span across all five sites), reflecting the typical 2–3 storey construction.

4.4 Spatial visualisation

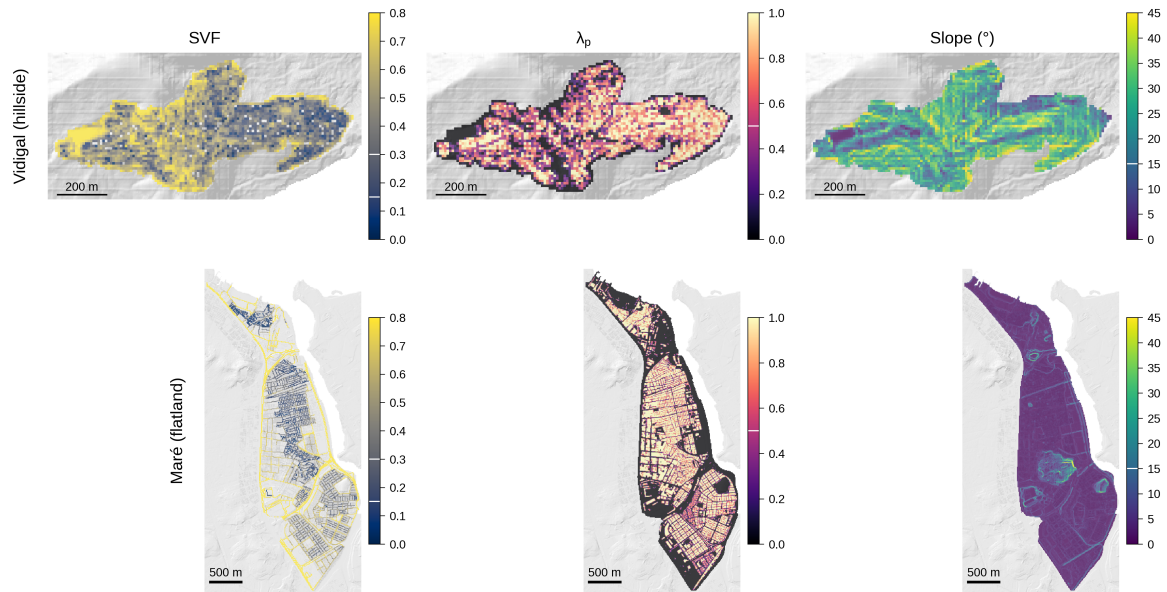


Figure 5. Spatial maps of SVF, λ_p , and slope for Vidigal (hillside) and Maré (flatland), with hillshade basemap and building outlines. Colour bars show indicator ranges; white contour lines on colourbars mark the stratification thresholds.

Notable observations:

- Vidigal's SVF map shows clear low-SVF corridors along the hillslope ridgelines where canopy density is highest (dark cyan in upper-left panel).
- Maré's λ_p is near-uniform at 0.7–1.0 across most of the settlement, reflecting the tight block structure of the original housing project layout.
- Slope maps confirm Vidigal's topography (green-yellow, 20–40°) vs Maré's flatness (purple, 0–3° except at engineered berms near the bay).

4.5 Indicator correlations

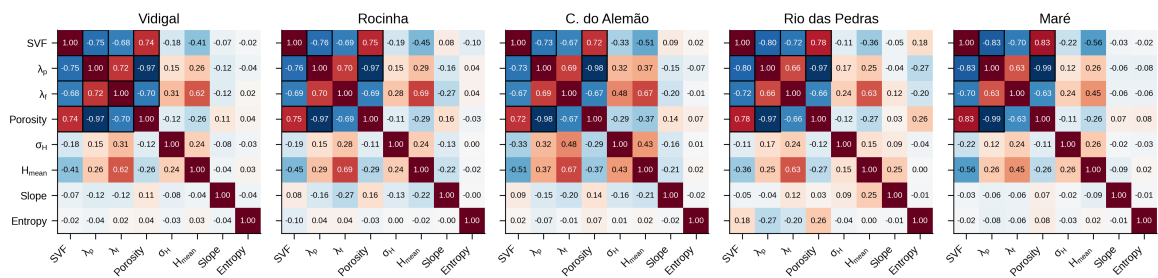


Figure S1. Pearson correlation matrices for the eight primary indicators at each site. Values are annotated in each cell; pairs with $|r| > 0.7$ are framed in black. Colour scale is fixed across all panels so patterns are directly comparable.

- **$\lambda_p \rightleftharpoons$ porosity** is the strongest correlation everywhere ($r \approx -0.95$), which motivates dropping porosity from the manuscript's distribution figure (Figure 2) to avoid redundancy.
- **$SVF \rightleftharpoons \lambda_p$** correlation is strong but not perfect ($r \approx -0.73$ to -0.83 across the five sites): variation in building *height* and arrangement breaks the $\lambda_p \rightarrow SVF$ link at fine scales. This is the structural-coupling finding developed in Section 5.
- **σ_H correlates weakly with H_mean** (per-site $r \approx 0.15$ – 0.43 , pooled ≈ 0.25). Earlier prose claimed $r \approx 0.5$; the current cross-site number is lower because the bimodal "regular block + tall outliers" pattern saturates in dense favela fabric.
- **Slope is nearly uncorrelated with morphology**, confirming that terrain and urban form are independent stratification axes — a necessary condition for the 12-stratum sampling scheme.

4.6 Resolution sensitivity

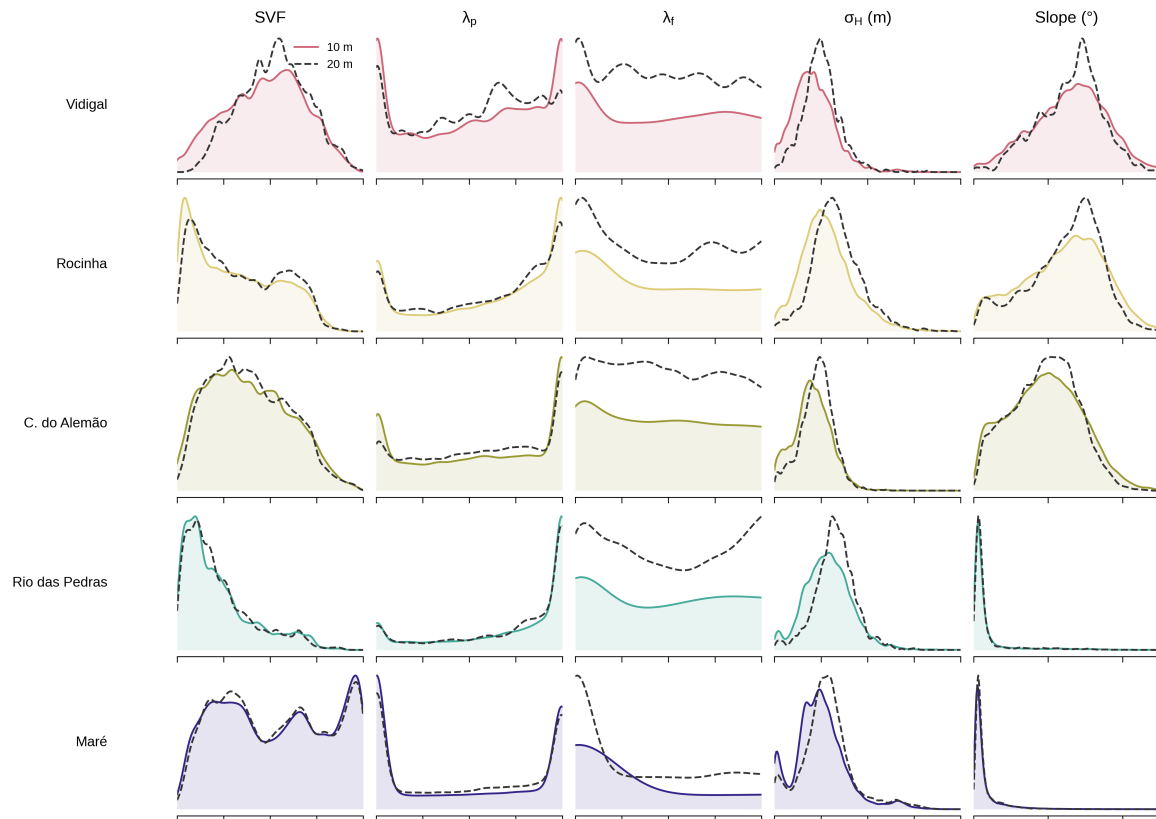


Figure S3. 10 m (solid coloured curves) vs 20 m (dashed black curves) morphometric distributions across all five sites and five indicators. The 10 m grid captures multimodal structure in λ_p and λ_f that 20 m aggregation smooths away — particularly visible in Complexo do Alemão and Rio das Pedras where the 20 m curves flatten features that clearly exist in the 10 m data.

This analysis justifies the 10 m resolution as the production grid:

- The key finding is preserved — distribution shape is similar — confirming that 10 m is not noise.
- But fine-scale multimodality relevant to CFD boundary conditions is only resolved at 10 m.
- Computational cost at 10 m is manageable (< 85 s per site).

Supplementary variants (Vidigal-only upscaled-vs-native scatter, and side-by-side difference maps) are archived in `outputs/paper_figures/exports/_variants/` and regenerable with `figS3_resolution_sensitivity.py --variants`.

5. Cross-Site Morphology

5.1 SVF- λ_p structural coupling

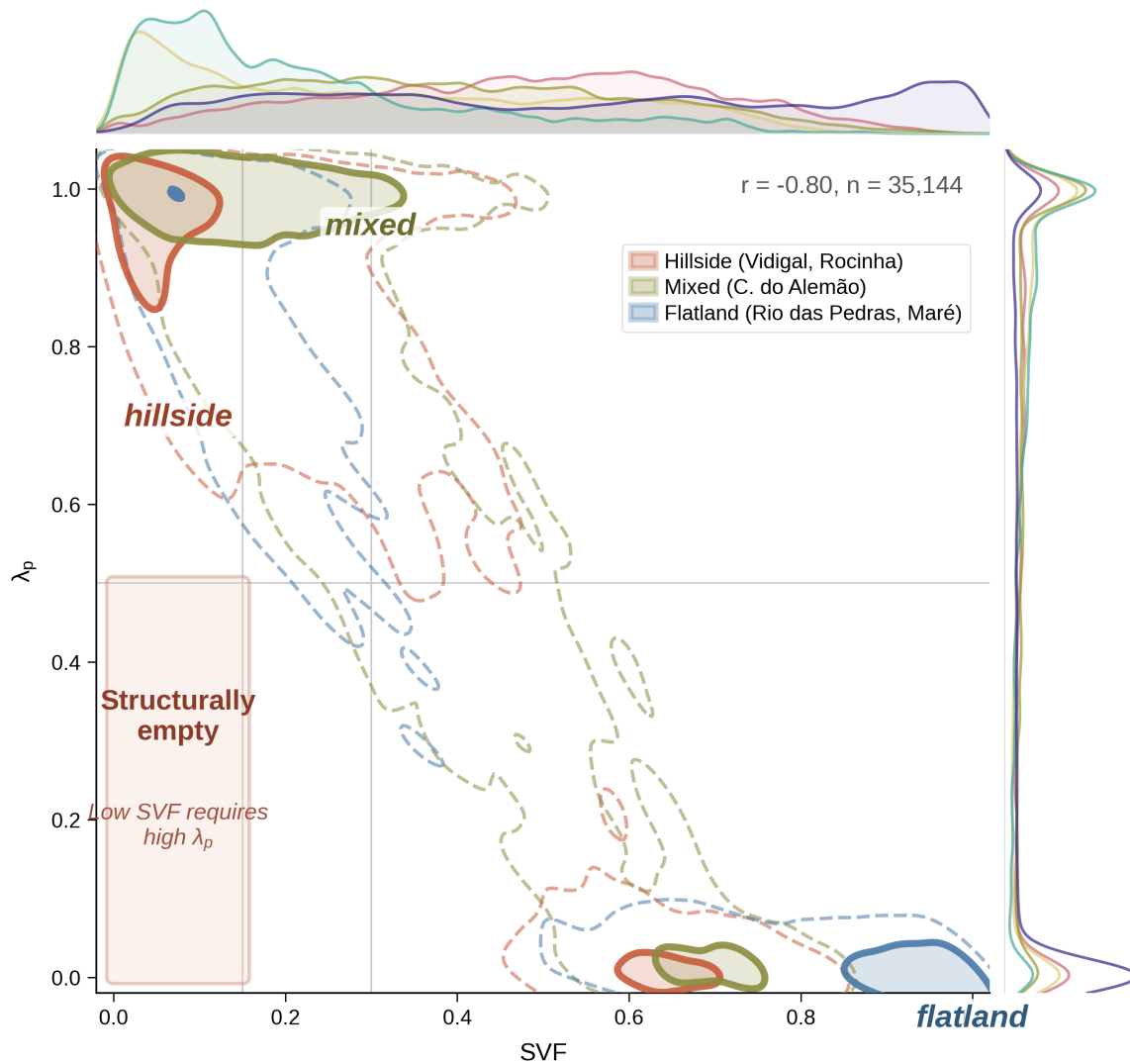


Figure 3. Central panel: 2D kernel density estimation of cells in (SVF, λ_p) space, grouped by typology. Solid contours enclose 50 % of each typology's cells; dashed contours enclose 90 %. Terracotta = hillside (Vidigal, Rocinha); olive-gold = mixed (Complexo do Alemão); steel blue = flatland (Rio das Pedras, Maré). Marginals show per-site SVF and λ_p distributions. The salmon-shaded region in the lower-left (SVF < 0.15, λ_p < 0.50) is **structurally empty** — only 102 cells in 35,144 (0.3 %) — because low sky view requires high plan density in realistic urban fabric. The strong negative correlation ($r = -0.80$ across all pooled cells) reflects this coupling. The three typologies occupy distinct but partially overlapping regions of the feature space: hillside cells cluster at higher λ_p for a given SVF, consistent with taller canopies; flatland cells spread to lower λ_p .

This figure is the central conceptual claim for the manuscript: the morphology of informal settlements occupies a constrained subset of the (SVF, λ_p) plane, and stratified CFD sampling must respect this constraint (empty strata are not allocation failures but physical impossibilities).

5.2 Typology summary statistics

Metric	Hillside (VDG+ROC)	Mixed (CDA)	Flatland (RdP+MAR)
Slope, median	25.7°	19.3°	1.5°
SVF, median	0.34	0.37	0.43
λ_p , median	0.71	0.62	0.54
λ_f _mean, median	3.43	2.20	3.12
σH , median	2.0 m	1.6 m	2.0 m

Source: medians computed from the concatenation of `outputs/{site}/morphometrics/grid/grid_metrics.gpkg` across the 12-strata eligible cells per site. Note the *Mixed* row reflects only Complexo do Alemão (which spans both flat and steep terrain — its slope median sits between the typology extremes); aggregating it with either Hillside or Flatland would compress that signal.

Two observations matter for the CFD design. First, **λ_p is uniformly high (0.54–0.71)** across all three typologies — informal urban form is denser than formal Rio fabric (typical λ_p in Copacabana ≈ 0.40), so the sampling allocation correctly prioritises high- λ_p strata. Second, **hillside and flatland sites carry comparable λ_f _mean (3.43 vs 3.12)**; the Mixed typology dips because Complexo do Alemão has the lowest building heights of the campaign (mean $H_{\text{mean}} = 5.3$ m). Wind-canopy drag will therefore not separate cleanly along the hillside $\leftrightarrow$ flatland axis; SVF and slope are the dominant discriminators that CFD will quantify.

5.3 SVF and solar access decouple on sloped terrain

On Vidigal's south face, terrain aspect adds 4–5 hr of solar variation at any SVF
winter sun in the southern hemisphere bypasses S-facing slopes

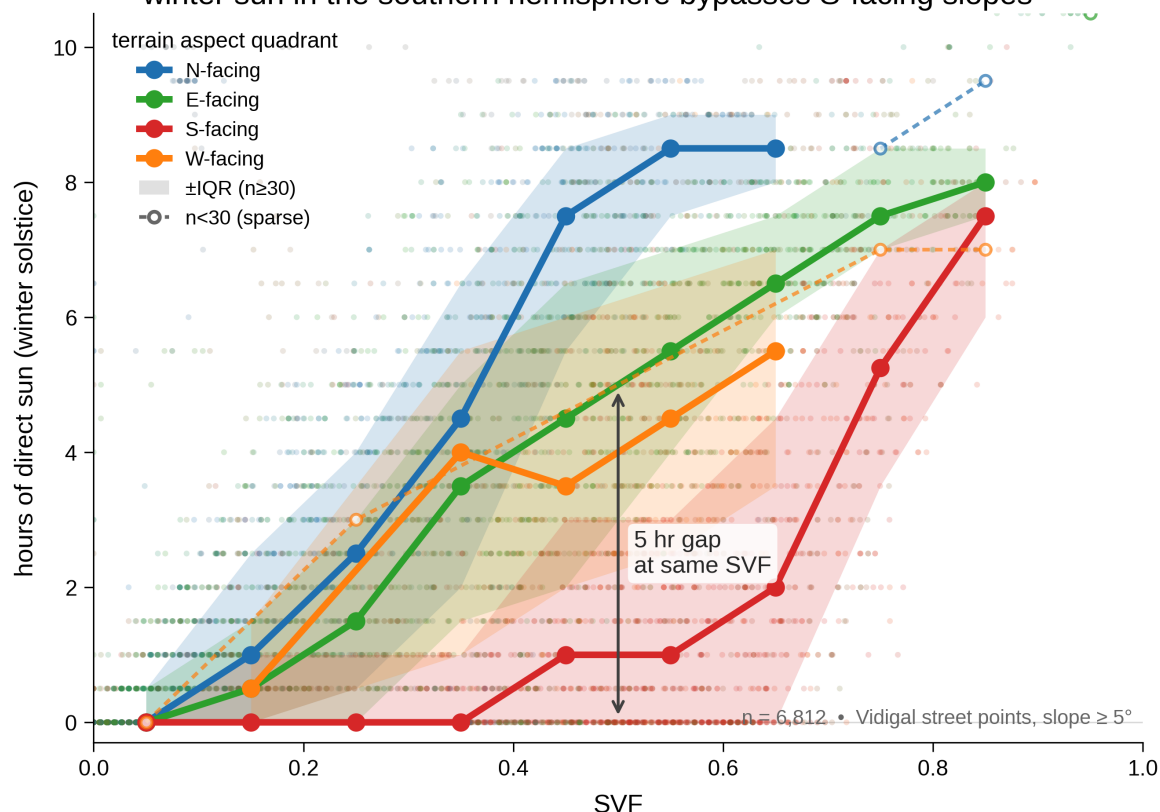


Figure 6. Per-quadrant median **winter-solstice** solar hours as a function of SVF, on Vidigal's sloped cells (slope $\geq 5^\circ$, $n = 6,812$ along-street points). The four traces — one per terrain-aspect quadrant (N / E / S / W) — do not collapse onto a single curve. At a fixed SVF ≈ 0.5 an N-facing cell receives ≈ 5 hours of direct sun on the winter solstice while an S-facing cell receives ≈ 0 ; the same is true at SVF ≈ 0.7 (8 hr vs 2 hr). Shaded bands show the inter-quartile range; trace segments where the SVF bin contains fewer than 30 cells are dashed (open markers) to flag the small-sample tails (W-quadrant beyond SVF ≈ 0.65). Underlay scatter is the full set of sample points coloured by aspect bearing, making the cloud's directional structure visible.

The headline scientific claim: **SVF and solar hours are not interchangeable on sloped terrain in the southern hemisphere.** SVF measures total sky visibility; the winter sun in Rio ($\sim 22^\circ\text{S}$) crosses the sky to the north, so a south-facing cell can have abundant sky visibility while pointing away from the sun's actual path. The W quadrant earns the most direct sun by catching the late-afternoon descent into the north-west horizon; the S quadrant is structurally shaded for the entire winter solstice day. This dissociation is the reason the §7 CFD regression uses aspect (via $(\sin \alpha, \cos \alpha)$) and aspect–wind alignment alongside SVF as distinct predictors, rather than collapsing them.

The same data is shown spatially in Figure S — terrain-aspect spatial expression, which maps four encodings of the Vidigal street-point set: terrain aspect (compass-coloured), SVF, solar hours, and the disagreement map SVF – solar (RdBu_r). Panel (d) of the supplementary figure is red-dominant across most of the favela: sky is open but the winter sun does not reach.

Quadrant means on Vidigal sloped cells (winter solstice, 30-min sampling, ray-cast against terrain + extruded LiDAR-height footprints): N = 3.32 h, E = 2.85 h, S = 1.70 h, W = 4.02 h, with N/E/S/W cell counts 1,363 / 2,595 / 2,488 / 366. Direct regression of solar hours on (slope, λ_p , $\sin \alpha$, $\cos \alpha$) reaches $R^2 = 0.246$ ($n = 6,876$, intercept 5.70 h, `aspect_cos` coefficient = +1.22 — i.e. moving from S- to N-facing under fixed slope and λ_p adds 2.4 h on average). Per-site SVF regressions on (slope, λ_p , $\sin \alpha$, $\cos \alpha$) are written to `outputs/{site}/morphometrics/aspect_regression.csv` (Vidigal SVF $R^2 = 0.599$) with per-quadrant summaries in `aspect_quadrant_summary.csv`. Cross-site signal: Vidigal sloped grid cells show SVF means N = 0.38 / E = 0.44 / S = 0.51 / W = 0.57 — the S/W cells have *more open sky* yet the figure shows them receiving *less winter sun*, which is exactly the dissociation point.

5.4 Seasonal envelope: the worst-case figure understates summer recovery

Vidigal — street-level solar envelope. Same points, same scale, three reference cases.

(a) §5.3 winter dissociation panel; (b) the figure year-round comfort & PV studies should use; (c) S-facing alleys recover in summer.

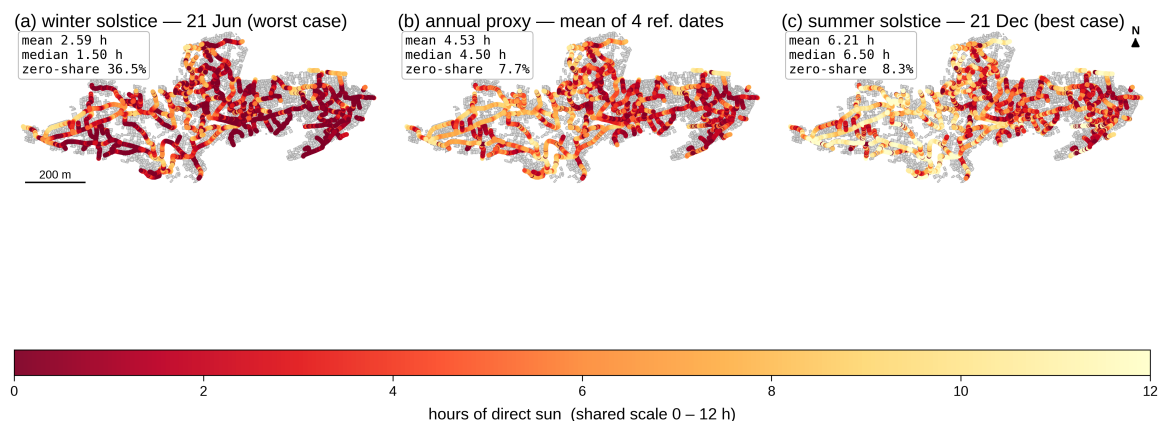


Figure 7. Three street-level solar maps on the same point set, same scale. (a) Winter solstice (worst case, mean 2.59 h, 36.5 % of points fully shaded). (b) Annual proxy: unweighted mean of four reference dates — winter and summer solstices plus March and September equinoxes (mean 4.53 h, 7.7 % fully shaded). (c) Summer solstice (best case, mean 6.21 h, 8.3 % fully shaded). Computed by `scripts/run_street_solar.py` (30-min sampling, 5 h–19 h window, pvlib sun positions, ray-cast against `scene.stl` = DTM + extruded

LiDAR footprints), with all four per-date arrays plus annual-mean columns written into a single `outputs/vidigal/morphometrics/svf/svf_streets_solar.gpkg`.

The seasonal swing on Vidigal is large enough that a winter-only analysis substantially understates summer access: 42.4 % of street cells gain ≥ 4 hours and 28.2 % gain ≥ 6 hours between winter and summer solstice, and 1,564 of the 2,510 winter-zero-hour cells (62 %) recover to ≥ 2 h in summer. Mean seasonal range across all street cells is 4.25 h. Under the WHO 2-hour daily-sunlight benchmark, 54.1 % of Vidigal's street cells fall short on the winter solstice but only 19.3 % on the summer solstice; **the year-round figure to use for comfort, photovoltaic, and health-outcome studies is the annual proxy (28.6 % below 2 h)**, not the winter worst case.

The standalone single-panel versions of Figure 7 (a/b/c) live at `docs/technical_report/figures/fig07a_solar_winter.png`, `fig07b_solar_annual.png`, and `fig07c_solar_summer.png`; they re-export from `outputs/paper_figures/fig07_solar_envelope_vidigal.py` and use the shared 0–12 h colour scale from that script for direct visual comparison.

6. CFD Patch Sampling

6.1 Sampling design

Each CFD simulation corresponds to one **analysis patch** of interest — a 100 m-diameter circle (radius 50 m, area $\approx 7\,854\text{ m}^2$) — embedded in a per-direction rectangular domain whose extents follow the Franke / COST 732 / Tominaga 2008 / Blocken 2015 wide-obstacle scheme (see §6.5 and `src/cfd_integration/rectangular_domain_v1.json`). The circular patch shape is independent of the rectangular mesh: it avoids corner artefacts in morphometric averaging, gives isotropic coverage independent of building-grid orientation, and is symmetric across all 8 wind-direction meshes. To characterise the morphological diversity of the five sites with a finite simulation budget, patches are allocated across **twelve stratification bins**:

- SVF: 3 bins (SVF < 0.15, $0.15 \leq \text{SVF} < 0.30$, SVF ≥ 0.30)
- Slope: 2 bins (< 15°, $\geq 15^\circ$)
- λ_p : 2 bins (< 0.5, ≥ 0.5)

giving $3 \times 2 \times 2 = 12$ strata, named e.g. `SVF1_SLP2_LP2` for SVF < 0.15 + slope $\geq 15^\circ$ + $\lambda_p \geq 0.5$.

6.2 Eligibility filter

A cell is an eligible patch centre if:

1. The CFD source-data envelope (rotated worst-case rectangle, half-diagonal $\sqrt{((10 \cdot H_{\max} + R_{\text{patch}})^2 + \text{lateral}^2)} + 50 \text{ m}$ per the v1 manifest) is fully covered by available building data. The original sampler used a 250 m-radius proxy at sampling time; the post-hoc v1 audit (`scripts/audit_rectangular_domain.py`) re-checks every campaign patch against the rectangular envelope using the 700 m extended-context buffer (§3.2, §6.5).
2. Within the 100 m-diameter circular analysis patch, building footprint coverage ($\sum \text{intersection area} \div \pi \cdot 50^2 \approx 7\,854 \text{ m}^2$) ≥ 0.5 . This also implicitly rejects patches clipped by the site boundary, where the missing area fails the coverage threshold.
3. All three stratification indicators are non-null.

Filter outcomes per site:

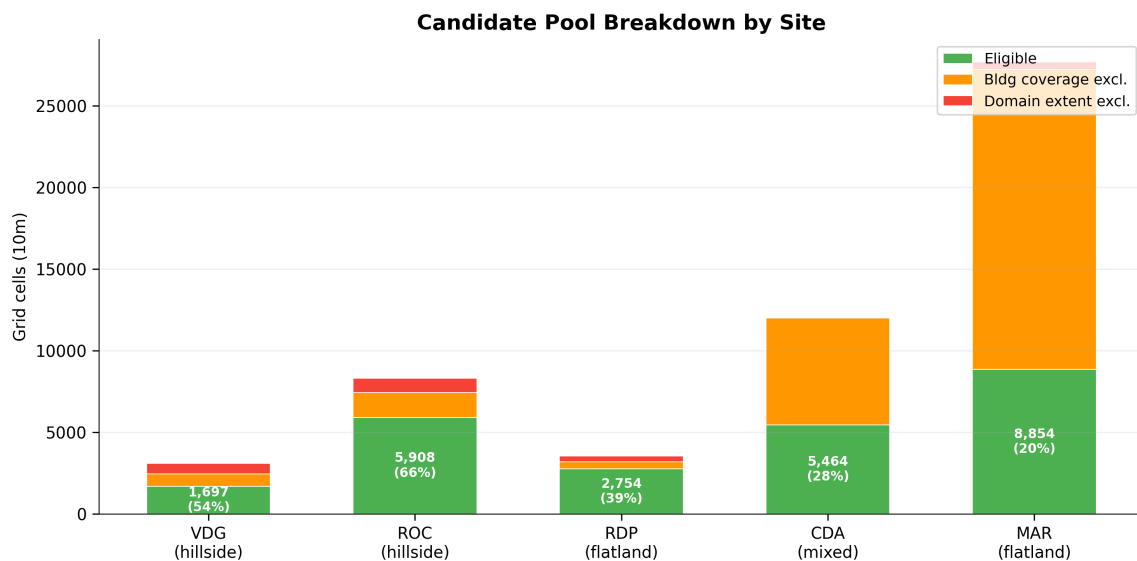


Figure (candidate pool). Stacked bars showing eligible cells (green) vs excluded cells per filter. Two exclusion sources: domain-coverage failure (red, dominated by edge cells in Vidigal and Rio das Pedras) and low-building-coverage failure (orange, dominant in Maré because of large residential block interiors without continuous footprint).

With the 300 m building-context extension, the domain-coverage exclusion drops from 55 % (no extension) to 0–20 % depending on settlement shape. The remaining exclusions are legitimate morphology: Maré's 20 % overall eligibility rate reflects its lower footprint coverage per 100 m patch, not a sampling artefact.

6.3 Selection algorithm

For each site, patches are chosen in two phases:

Phase 1 — Pilot (12–15 patches per site). Within each non-empty stratum, at least one patch is allocated; an additional bonus is given to the three most populated strata. Patches are chosen by **greedy maximin geographic distance** starting from the two most distant eligible cells; subsequent picks maximise minimum distance to already- selected patches (and to patches of other strata selected earlier, so that the cross-stratum spacing constraint is respected). Minimum spacing: 80 m between patch centres. With 100 m-diameter patches, two patches at the minimum centre-to-centre distance overlap in a lens of $\approx 817 \text{ m}^2$ ($\approx 10.4 \%$ of each patch area); this residual overlap is accepted because the patches are evaluated independently in separate CFD simulations and no joint statistics are computed over overlapping footprints.

Phase 2 — Campaign (incremental to 22–25 patches per site). Pilot patches are held fixed and treated as immovable seeds; additional patches are chosen with **SVF-priority weighting** to oversample the health-relevant low-SVF strata:

- SVF1 (< 0.15) $\rightarrow \times 2.0$ multiplier
- SVF2 ($0.15\text{--}0.30$) $\rightarrow \times 1.0$
- SVF3 (≥ 0.30) $\rightarrow \times 0.8$

The effective allocation is proportional to the weighted eligible count per stratum, subject to per-stratum minimums (≥ 1 patch if any cells exist) and the 80 m spacing constraint against the union of pilot and newly-selected patches.

Implementations: `scripts/run_pilot_sampling.py` (Phase 1), `scripts/run_campaign_sampling.py` (Phase 2), `scripts/build_extended_context.py` (upstream).

6.4 Allocation results

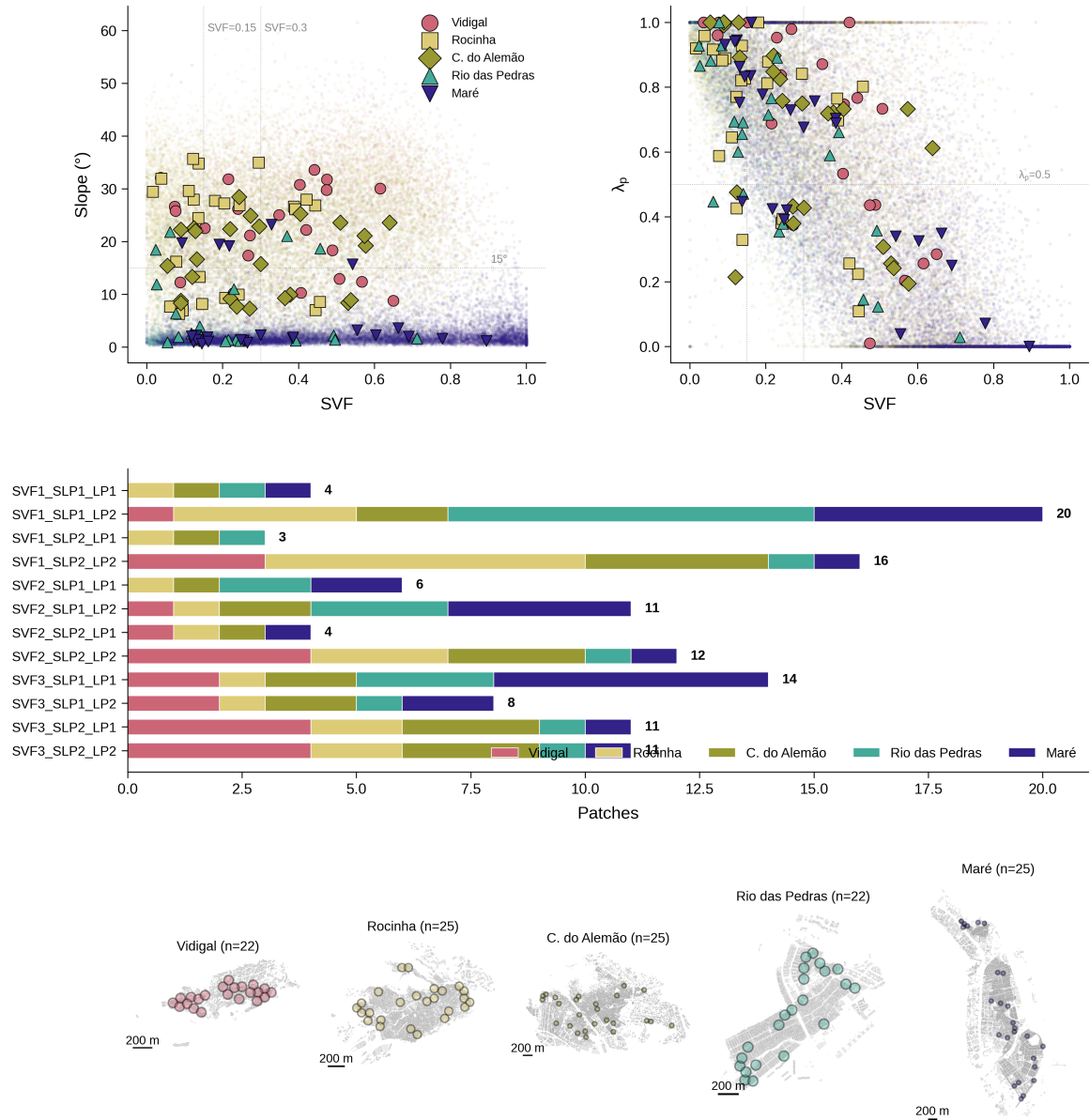


Figure 4. (a–b) Feature-space coverage of the 119 selected patches over the full grid-cell cloud for (SVF, slope) and (SVF, λ_p) axes. Dotted lines mark stratum boundaries. (c) Horizontal bar chart of patches per stratum, with segments coloured by site; numeric labels give the stratum total. (d) Per-site maps showing analysis-patch locations as coloured 100 m-diameter circles over grey building footprints. All five sites shown at consistent visual scale with 200 m scale bars.

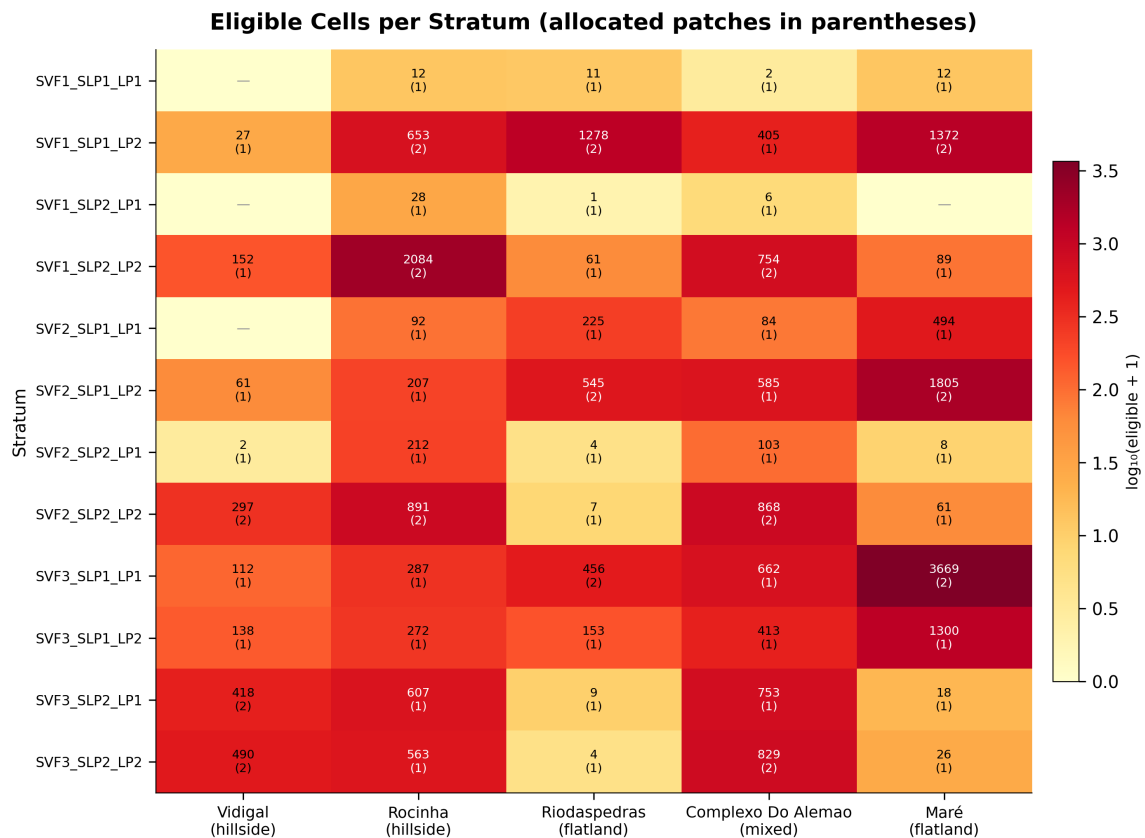


Figure (strata heatmap). Matrix of eligible cell counts ($\log_{10}$ scale) and allocated patches per stratum per site. Cells with no eligible candidates are marked with an em-dash. This makes the partial-empty structure explicit: three strata are empty at one or more hillside sites (SVF1_SLP1_LP1, SVF1_SLP2_LP1, SVF2_SLP1_LP1) because Vidigal's narrow hillside settlement has no flat cells with sparse footprints.

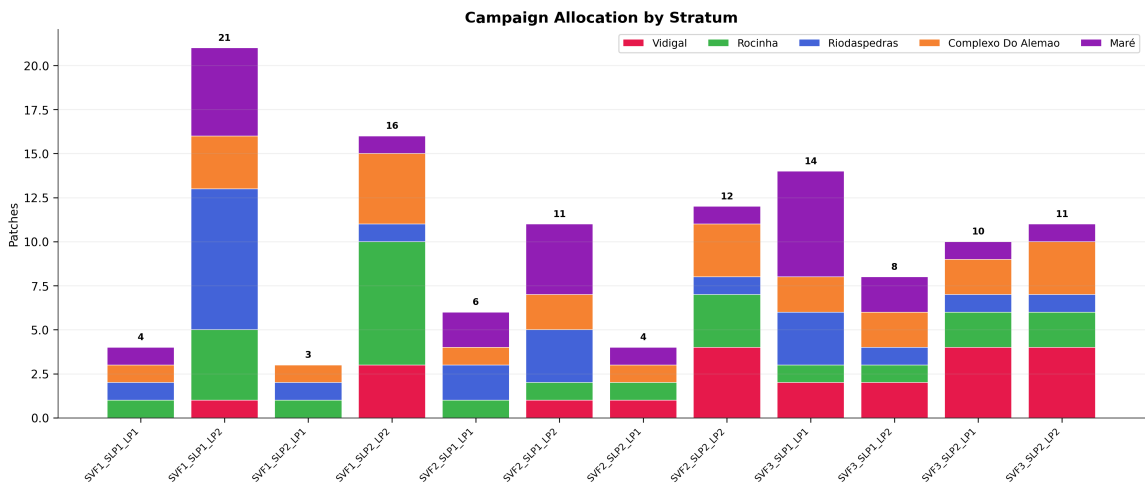


Figure (campaign allocation). Total patches per stratum in the final 119-patch campaign, segments coloured by site. SVF1_SLP1_LP2 (low sky view, flat, dense) has the highest patch count (20) after SVF-priority weighting — this is the stratum most relevant to health outcomes at the flatland sites and was intentionally oversampled.

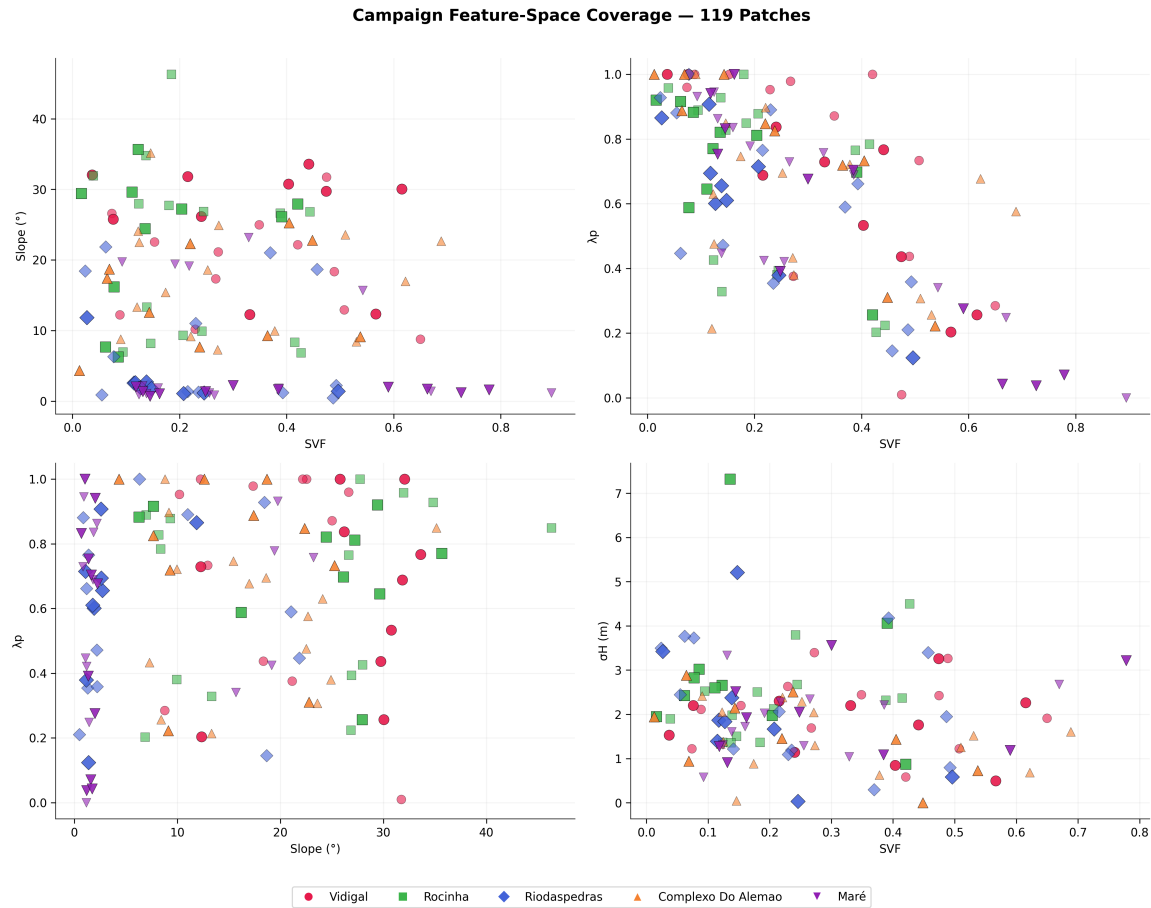


Figure (cross-site feature space). All 119 patches overlaid on all five sites' grid-cell clouds across four feature-space projections (SVF vs slope, SVF vs λ_p , slope vs λ_p , SVF vs σ_H). Site colours match throughout. Verifies that sampled patches span the full morphological envelope of the five sites without clustering in any projection.

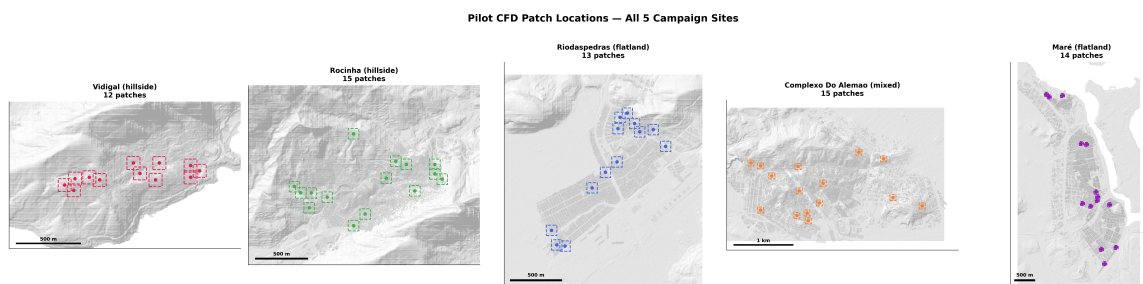


Figure (all sites overview). 5-panel map of final patch locations per site, each with hillshade basemap and patch outlines in site colour.

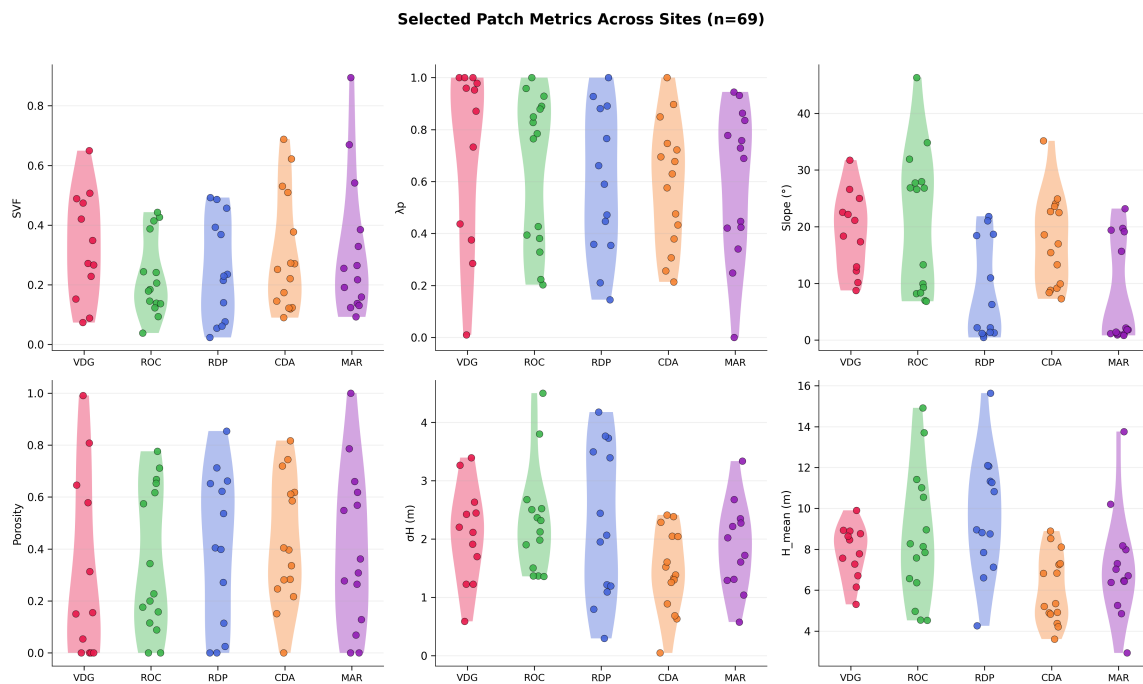


Figure (patch metrics). Violin plots of morphometric values at the 119 selected patches, grouped by site. Confirms that the selection does not over-represent or under-represent any single site's typical conditions: each site's patch distribution spans its full grid-cell distribution for each indicator.

Final per-site allocation:

Site	Pilot	Campaign add	Total	Shortfall
Vidigal	12	10	22	0
Rocinha	15	10	25	0
Rio das Pedras	13	9	22	-1
Complexo do Alemão	15	10	25	0
Maré	14	11	25	0
TOTAL	69	50	119	-1

Rio das Pedras has a one-patch shortfall against its 23-patch target: the SVF2_SLP2_LP2 stratum has only 7 eligible cells, all within 80 m of existing pilot patches, so no additional candidate could be placed without violating the minimum-spacing constraint. This is a genuine morphological limitation (the stratum is physically scarce) and was accepted rather than relaxing the spacing.

6.5 Rectangular per-direction domain compliance

The CFD methodology adopted in May 2026 is rectangular per-direction (one `blockMesh` per patch × wind-direction pair) per Franke et al. (2007) / COST 732 / Tominaga et al. (2008) AIJ / Blocken (2015). The canonical rule set is `src/cfd_integration/rectangular_domain_v1.json`. Domain dimensions per patch:

```

upstream    = 5 · H_max + R_patch
downstream  = 15 · H_max + R_patch
lateral     = max(5 · H_max + R_patch, 5 · W_patch)  each side
top         = 5 · H_max

```

with $R_{\text{patch}} = 50$ m (analysis-circle radius) and $W_{\text{patch}} = 100$ m (diameter). Because all 119 patches have $H_{\text{max}} < 90$ m, the lateral floor of $5 \cdot W_{\text{patch}} = 500$ m dominates, and the silhouette-envelope blockage $D \cdot H_{\text{max}} / (2 \cdot \text{lateral} \cdot \text{top})$ collapses to a *uniform* 2 % across the campaign — independent of H_{max} , well under the 5 % AIJ gate (Tominaga 2008). All 119 patches pass blockage at this gate.

The rotated-rectangle source-data envelope (a single radius per patch that covers all 8 wind-direction meshes) ranges 565–646 m across the campaign, with the maximum at RDP-P15 ($H_{\text{max_analysis}} = 27.3$ m). The extended-context layer was extended from 300 m to 700 m on all five sites (`buildings_extended_700m.gpkg` / `dtm_extended_700m.tif`) in the same migration. After the buffer extension all 119 patches are eligible — *no patch was dropped or replaced*, the campaign list is intact.

The earlier cylindrical-domain rule (radius 250 m, Blocken minimum fetch $5 \times H_{\text{max}}$) is deprecated. The indicator column `blocken_radius_required` was dropped from both `outputs/{site}/sampling_cfd/campaign_sampling/campaign_patches.csv` and `outputs/{site}/cfd_analysis/per_patch_indicators.csv` and replaced by a per-row `domain*_m` set (upstream, downstream, lateral, top), `domain_blockage_ratio` and `domain_blockage_ok`, `source_data_required_m`, `source_data_extent_m`, and `eligible`. Real per-direction frontal area density λ_F is also recorded per patch (8 directions + mean + max), but is *not* the blockage gating quantity. $\lambda_F = \Sigma(\text{projected_facade} \times \text{height}) / \text{disk_area}$ is the canopy-parameterisation form (Tominaga 2008 §3) — sum of all building facades, no shadowing — and routinely exceeds 1.0 in dense favela patches (campaign median λ_F max = 1.32, p90 = 2.10, max = 2.96 at ROC-P01). The CFD blockage gate uses the AIJ benchmark *silhouette envelope* $D \cdot H_{\text{max}}$ instead: the worst-case projected obstacle area in any wind direction, bounded by treating the analysis circle as a solid block. λ_F is reported alongside for the §4 morphometric tables and downstream urban-canopy analyses.

7. CFD Integration Pipeline

7.1 Overview

The code to ingest CFD results, aggregate them onto the 10 m morphometric grid, and combine directional simulations into an annualised ensemble is implemented in `src/cfd_integration/` (~1,100 lines across five modules: `schema.py`, `io.py`, `aggregate.py`, `metrics.py`, `weighting.py`) and covered by **63 passing unit tests**.

Input contract (documented fully in `src/cfd_integration/README.md`, provided to the CFD agent as the delivery specification):

```
data/{site}/cfd_results/{patch_id}/{wind_direction}/
  sample_points.csv      # required – 15k rows at z = 1.5 m, 2 m spacing
  summary.json           # required – simulation metadata
  field.vtu              # optional – full 3D field
```

Plus `data/{site}/wind_rose.json` per site (now built from measured INMET BDMEP / Iowa ASOS records; see §2.3 and Figure S5).

7.2 Module architecture

- **`schema.py`** — dataclasses for sample points, per-simulation metadata, campaign aggregates, wind roses; defines the 8 cardinal directions and runtime validators.
- **`io.py`** — CSV + JSON loading (primary path), optional VTU loading via `pyvista` (fallback), directory traversal for per-site campaigns.
- **`aggregate.py`** — spatial aggregation onto the morphometric grid. Primary function `aggregate_to_grid()` maps CFD samples onto individual 10 m cells whose centroid falls inside the 100 m-diameter circular analysis patch (the scientifically defensible zone per Tominaga 2008 / Blocken 2015). `aggregate_to_domain()` is a supplementary function that includes the rest of the rectangular domain (out to the per-direction blockMesh boundary) for robustness checks.
- **`metrics.py`** — health-relevant scalar quantities:
 - `velocity_magnitude()` — $\sqrt{U^2 + V^2 + W^2}$
 - `stagnation_fraction()` — fraction of samples with $|U| < \text{threshold}$ (default 0.5 m/s, the commonly cited calm-air limit)
 - `turbulent_intensity()` — $TI = \sqrt{(2/3 \cdot TKE)} / U_{ref}$
 - `ach()` — air change rate using the standard urban-climate formulation $ACH = 3600 \times \langle |U| \rangle / L$
 - `low_wind_percentile()` — worst-case stagnation (default p10)
 - `canyon_ventilation_efficiency()` — $\langle |U| \rangle / U_{ref}$

- **weighting.py** — wind-rose-weighted combination of 8 directional simulations into an annualised metric. Supports three weighting modes: uniform (1/8 each), frequency-only ($\propto f_{\text{dir}}$), and frequency $\times$ speed ($\propto f_{\text{dir}} \cdot U_{\text{dir}}$, which emphasises directions contributing more wind energy). `worst_case_direction()` finds the direction maximising any chosen metric (e.g. the direction producing most stagnant flow).

7.3 CFD agent specification

The full delivery contract (CSV schema, JSON schema, directory layout, methodology requirements for turbulence model, boundary conditions, meshing, post-processing, and convergence targets) is documented in `src/cfd_integration/README.md`. Key decisions baked into the contract:

- **Simulation domain:** rectangular per-direction (one `blockMesh` per patch $\times$ wind-direction). Per-patch extents `5/15/5/5 · H_max + R` with `lateral = max(5·H+R, 5·W) = 500 m` for our 100 m patches. Top = `5 · H_max`. See `rectangular_domain_v1.json` for the full rule set. Pedestrian sampling only valid within the central 100 m- diameter circular analysis patch.
- **Turbulence model:** $k-\omega$ SST (standard RANS for ABL flows at this scale).
- **Inlet:** logarithmic velocity profile, suburban roughness class ($z_0 \approx 0.03$ m), turbulence intensity ≈ 0.15 at 10 m.
- **Simulations per site:** 8 cardinal directions, sharing one mesh with rotated inlet BC. 119 patches $\times$ 8 directions = 952 total simulations for the full campaign.
- **Sampling:** horizontal slice at $z = 1.5$ m, 2 m in-plane spacing, OpenFOAM `sample` utility, delivered as CSV.
- **Recommended validation patch: MAR-P07** (Maré, flat, 12.2 m max height, 1,200 buildings in domain) — the simplest end-to-end test case to validate the full pipeline before scaling.

7.4 End-to-end usage (post-CFD)

```

from src.cfd_integration import (
    load_campaign_results, aggregate_to_grid, weighted_by_wind_rose,
)

campaign = load_campaign_results("vidigal") # reads data/vidigal/cfd_results/
for patch_id in campaign.patch_ids():
    per_direction = {
        dir_: aggregate_to_patch(campaign.patches[patch_id][dir_],
                                patch_center_xy)
        for dir_ in campaign.directions_for(patch_id)
    }
    annual = weighted_by_wind_rose(per_direction, campaign.wind_rose,
                                   weight_by="freq_speed")
    # annual = {annual_U_mean, annual_stagnation_frac, annual_ach, ...}

```

`scripts/analyze_cfd_results.py` runs the four-step chain (annualise per patch → annualise per cell → predictor OLS → fig 5 wind panel) and writes `predictor_regression.csv`. Predictors are SVF, λ_p , slope, σ_H , and three terrain-aspect covariates: `aspect_sin`, `aspect_cos` (the orthogonal pair for the circular aspect, computed per patch as the circular mean of `aspect_deg` over the 100 m analysis disk), and `aspect_wind_alignment` = $\cos(\text{aspect} - \text{dominant-wind-direction})$. The dominant wind direction is the frequency-weighted circular mean of the per-site wind rose. On the synthetic dataset that exercises this chain end-to-end the aspect coefficients are at noise level (the synthetic generator does not encode aspect into `U_mag`); they will become non-trivial once real CFD returns the directional flow field.

8. Repository Structure

Canonical output layout (as of 2026-04-13):

```

outputs/
├─ {site}/
│  ├─ morphometrics/          # SVF, per-building metrics, 10m grid,
│  │  ├─ svf/                #   figures, report
│  │  ├─ buildings/
│  │  ├─ grid/
│  │  ├─ figures/
│  │  └─ report/
│  ├─ morphometrics_20m/      # 20m grid (for resolution sensitivity)
│  └─ sampling_cfd/
│     ├─ pilot_sampling/      # 12–15 patches
│     └─ campaign_sampling/   # 22–25 patches (pilot + new)
├─ comparative/              # cross-site analysis
│  ├─ pilot_summary/          # pilot-phase cross-site comparison
│  ├─ final_allocation/       # 119-patch campaign summary
│  └─ audits/                 # topology audits
├─ paper_figures/            # manuscript figures
│  ├─ fig_style.py            # shared style module
│  ├─ fig01_study_sites.py
│  ├─ fig02_morphometric_distributions.py
│  ├─ fig03_svf_lambda_coupling.py
│  ├─ fig04_sampling_design.py
│  ├─ fig05_morphometric_maps.py
│  ├─ figS1_correlation_matrices.py
│  ├─ figS2_context_extension.py
│  ├─ figS3_resolution_sensitivity.py
│  ├─ figS4_patch_thumbnails.py # outputs not currently published in the TR
│  ├─ figS5_wind_roses.py
│  ├─ README.md
│  └─ exports/                # rendered PNG + SVG

```

Source modules:

```
src/
├─ config.py           # paths, CRS, filtering thresholds
├─ cartography.py      # scale bars, north arrows, UTM formatting
├─ morphometry/       # grid computation
│   ├─ grid.py
│   ├─ indicators.py
│   ├─ audit.py
│   ├─ figures.py
│   └─ report.py
├─ svf_v2/            # SVF engine (Tregenza-145 ray-cast)
│   ├─ compute.py      # per-point and parallel raycast drivers
│   ├─ scene.py        # 3D scene from DTM + footprints
│   ├─ sampling.py     # passageway / street-centreline samplers
│   ├─ paths.py        # per-site path registry
│   ├─ io.py           # per-cell aggregation + raster I/O
│   ├─ utils.py
│   ├─ facades.py      # façade-side sampling
│   └─ visualize.py
├─ solar/             # solar access (phase 4)
└─ cfd_integration/   # CFD ingestion (phase 7)
    ├─ schema.py
    ├─ io.py
    ├─ aggregate.py
    ├─ metrics.py
    ├─ weighting.py
    └─ README.md      # CFD agent specification
```

Key scripts:

Script	Purpose
run_morphometric_audit.py	Per-site morphometric grid + SVF audit + report
build_extended_context.py	Site + city-wide building and DTM merger
run_pilot_sampling.py	12-strata pilot batch (12–15 patches per site)
run_campaign_sampling.py	Campaign allocation (SVF-priority, target-total)
build_wind_rose.py	INMET → wind_rose.json ingestion



9. Validation Summary

Check	Status	Evidence
Building geometry validity	PASS (10 repaired, all sites clean)	outputs/comparative/audits/rocinha_topology_audit.md
CRS consistency	PASS	all inputs verified EPSG:31983 on load
SVF passageway aggregation (no centroid artefacts)	PASS	diagnostic in <code>src/morphometry/grid.py::_aggregate_svf_to_grid</code>
DTM sentinel value handling	PASS (post-fix)	sentinel = -9999 pixels masked in extended-context merge per <code>scripts/build_extended_context.py</code>
Extended context eligibility recovery	PASS	Vidigal eligible cells go from ~0.45 to 0.80 of grid with the 300 m extension; see <code>outputs/comparative/pilot_summary/candidate_pool_comparison.csv</code>
Rectangular-domain v1 eligibility (all 119 patches)	PASS	<code>eligible = true</code> in every row of <code>outputs/comparative/cfd_methodology/audit_v1.csv</code> ; uniform 2 % silhouette blockage (well under 5 % AIJ); source-data envelope 565–646 m covered by <code>buildings_extended_700m.gpkg</code> on all 5 sites
Minimum inter-patch spacing (80 m)	PASS	realised minimum: 80–85 m across sites
Resolution sensitivity (10 m justified)	PASS	Figure S3; 10 m captures features 20 m loses
12-strata coverage (≥ 1 site per stratum)	PASS	all 12 strata non-empty when pooled
CFD integration unit tests	PASS	63 / 63 tests

10. Known Limitations

10.1 Neutral stability is assumed

All five `wind_rose.json` files now carry measured hourly observations (`quality_flag: "measured"`, 2015–2024 window, $n = 64,088\text{--}89,439$; see §2.3). The methodological simplification that remains is the assumption of neutral atmospheric stability across all directions and seasons. Stability classes are not separated; CFD inflow uses the neutral log-law profile. In practice this is conservative for the dispersion-relevant low-wind regimes that dominate Rio das Pedras and Complexo do Alemão (calm fractions 46 % and 33 %), and a known limitation for the more stably stratified nocturnal hours.

10.2 CFD results not yet integrated

`src/cfd_integration/` is tested but has not yet processed real simulation data. Any assumptions about sample-point density, column naming quirks, or edge cases in OpenFOAM output will only surface at first ingestion. The first pilot patch (VDG-P07) is in flight; running the test suite plus `cfd-results-ingestor` agent against the returned results will catch most issues.

10.3 SVF cross-validation against UMEP

(Limitation closed 2026-04-29 for Vidigal; height-matched at $z = 1.5$ m for Vidigal and Maré 2026-04-30; extended to all 5 sites 2026-05-01.) The Tregenza 145-patch engine was cross-validated against UMEP's shadow-casting SVF (`svfForProcessing153`) at $z = 1.5$ m after lowering building heights by 1.5 m to height-match the two engines:

Site	n	r ²	slope	RMSE	bias
Vidigal	2,510	0.68	0.96	0.12	+0.01
Rocinha	5,646	0.81	1.01	0.14	+0.09
Rio das Pedras	1,503	0.93	1.25	0.11	+0.08
Complexo do Alemão	4,838	0.76	0.92	0.17	+0.14
Maré	9,516	0.94	0.97	0.12	+0.09

Four of five sites agree at slope within $\pm 8\%$ of unity (Vidigal, Rocinha, Complexo do Alemão, Maré). Rio das Pedras at slope = 1.25 reflects its small valid grid ($n = 1,503$) concentrated in the 0.3–0.5 SVF band where the 153-patch shadow-cast and 145-patch ray-cast integrations diverge most. Both engines are defensible operational definitions of SVF; the 5-site slope- ~ 1 cluster establishes that the IVF Tregenza-145 ray-cast engine is consistent with an independent benchmark across the full morphological range represented in the campaign. See §4 cross-validation table for per-site interpretation and `outputs/{site}/morphometrics/svf/umep_validation/scatter.png` (S6 Vidigal, S7 Maré, S8 Rocinha, S9 Rio das Pedras, S10 Complexo do Alemão) for per-cell scatters.

10.4 Resolution sensitivity is 10 m vs 20 m only

Finer grids (5 m, 2 m) would be prohibitively expensive at site scale but warrant a spot-check for the CFD patches specifically. Not a priority for the current milestone.

10.5 Cidade de Deus excluded

A data integrity issue in the CDD building footprints (geometry inconsistencies; see project memory `project_cdd_data_bug`) leaves CDD outside the 5-site campaign. This is a final decision for the current campaign cycle: the 119-patch

allocation, OpenFOAM submission, and downstream analysis all assume 5 sites. Re-onboarding CDD is out of scope until the building data is reprocessed upstream of this repository.

10.6 Direct ground-solar measurements only on Vidigal

`scripts/run_street_solar.py` produces a four-date seasonal envelope (winter and summer solstices, two equinoxes, 30-min sampling) for Vidigal at `outputs/vidigal/morphometrics/svf/svf_streets_solar.gpkg`. The `gpkg` carries `solar_hours_winter / summer / equinox_mar / equinox_sep / annual` plus matched irradiance columns; §5.3 and §5.4 both consume it. The legacy `scripts/compute_solar_access.py` remains in the repo but is superseded — its single-date 60-min output was replaced by the seasonal runner on 2026-05-07.

The remaining four sites (Rocinha, Complexo do Alemão, Rio das Pedras, Maré) lean on SVF as the per-cell sky-exposure proxy. The seasonal runner is site-agnostic: only Vidigal and Maré currently have a pre-built `scene.stl`, so re-using it on the other three sites first requires running `python scripts/run_svf_v2.py --area <site> --mode streets` to materialise their scene meshes (~15–30 min per site), after which `python scripts/run_street_solar.py --site <site>` adds the seasonal envelope (~2 min per site at 16 cores). Total cross-site catch-up: ~1.5 hours of compute, no methodology changes.

Until that runs, the §5.4 Figure 7 narrative is Vidigal-only; §5.3 already states the SVF ↔ solar dissociation in cross-site terms via the shared aspect predictor. Cross-site solar coverage is the next paper-figure follow-up, not a methodology gap.

11. Next Steps

In this repository:



Wind ingestion — measured roses for all 5 sites (completed 2026-04-27, see §2.3)



Build the result-side analysis pipeline (`scripts/analyze_cfd_results.py`) end-to-end on synthetic CFD data so the chain is exercised before VDG-P07 returns (completed 2026-04-29; smoke-tested across all 5 sites 2026-04-30: 359–405 covered cells per site, predictor signs consistent with morphology). Producer ↔ validator loop closed via `cfd-results-ingestor` against the

synthetic outputs; both surfaced WARNs fixed in `scripts/generate_synthetic_cfd_results.py` so future synthetic runs hit PASS.



Cross-validate SVF against UMEP shadow-cast reference at pedestrian height across all 5 sites (closed 2026-05-01; slopes 0.92–1.25 with four of five within $\pm 8\%$ of unity; see §10.3 and §4).



Migrate the campaign domain methodology from cylindrical (radius 250 m) to rectangular per-direction (Franke / COST 732 / Blocken 2015 wide-obstacle scheme); audit all 119 patches against the new rule set; extend `buildings_extended_*` and `dtm_extended_*` from 300 m to 700 m on all five sites; drop the deprecated `blocken_radius_required` indicator column; publish `src/cfd_integration/rectangular_domain_v1.json` as the canonical contract (closed 2026-05-08, see §6.5; 119/119 eligible after the buffer extension, no patches dropped).



Optional: 20 m grids already exist; generate the variant Fig S3 forms (`--variants` flag) if reviewers request the scatter or difference-map views



Draft the Nature Cities manuscript (separate from this technical report) once the result-side pipeline is operational

In the CFD repository:



Implement OpenFOAM case from per-patch exports (template in `outputs/{site}/sampling_cfd/campaign_sampling/patches/{id}/`) using the v1 rectangular domain rule set



~~Validate pipeline on VDG-P07~~ — VDG-P07 cylindrical pilot killed 2026-05-08 in favour of selecting fresh pilots from the v1 rectangular campaign (see §6.5); pilot picks deferred to the Airflow side



Mesh convergence study (one v1 pilot per morphology stratum)



Run full 952-simulation campaign



Post-process to `sample_points.csv` + `summary.json` per simulation

After CFD results land here. All of the machinery for these steps was built and synthetic-validated in commits `c9d25d6` (pipeline) and `0851b5e` (synthetic-generator self-validation); each box flips when the same code path is re-run against real Airflow output.



Verify first patch ingests correctly via `cfid-results-ingestor` (validator already accepts both IVF-native CSV and Airflow-native parquet layouts; tested against synthetic 160/160 cleanly)



Annualise via wind-rose weighting (`src.cfd_integration.weighting`) — implemented; default weight is `freq_speed`



Map CFD metrics onto the 10 m grid (nearest-patch v0) — implemented; covered cells 359–405 per site under synthetic



Extend Figure 5 to include wind-velocity panel (`outputs/{site}/cfd_analysis/figures/fig5_wind_panel.png` generated under synthetic; awaits real numbers for the report)



Statistical analysis: SVF/ λ p/terrain as predictors of ACH and stagnation — predictor regression CSV generated per-site under synthetic with expected sign structure



Health-outcome linkage

12. Reproducibility

Every numerical claim, table, and figure in this report can be regenerated from the committed scripts. This section is the index.

12.1 Environment

```
git clone https://github.com/thrmnn/MorphoFavela.git && cd MorphoFavela
conda create -n IVF python=3.11 && conda activate IVF
pip install -e ".[dev]" # add ".[gpu]" for the optional GPU SVF stack
```

GDAL / GEOS native libraries must be present (`apt install libgdal-dev` on Linux, `brew install gdal` on macOS) before `pip install`.

12.2 Smoke test (≤ 2 min on a fresh clone)

```
pytest tests/ -m "not integration" -q --tb=short
# → 508 tests pass; 69 integration tests deselected
```

If this passes, the codebase is loaded correctly. If GDAL or pyvista fail to import, see `docs/onboarding/local_setup.md`.

12.3 Per-site pipeline

Substitute any of `vidigal` | `rocinha` | `riodaspedras` | `complexo_do_alemao` | `maré` for `<site>`. Inputs must be in place under `data/<site>/` per `data/README.md`.

```
# Stage 1: extended building + DTM context – buffer must cover the
# v1 rectangular CFD source-data envelope (max 646 m). Use 700 m.
python scripts/build_extended_context.py --area <site> --buffer 700

# Stage 2: morphometric grid + SVF audit + per-site report
python scripts/run_morphometric_audit.py --area <site>
# Outputs → outputs/<site>/morphometrics/{grid,svf,figures,report}/

# Stage 3: CFD patch sampling
python scripts/run_pilot_sampling.py --site <site> \
  --buildings data/<site>/buildings_extended_700m.gpkg \
  --dtm data/<site>/dtm_extended_700m.tif
python scripts/run_campaign_sampling.py
# Outputs → outputs/<site>/sampling_cfd/campaign_sampling/patches/<PATCH_ID>/

# Stage 4: rectangular-domain v1 audit + indicator migration (cross-site,
# run once after all per-site sampling is complete)
python scripts/audit_rectangular_domain.py
python scripts/migrate_indicators_rectangular_v1.py --apply
# Outputs → outputs/comparative/cfd_methodology/audit_v1.csv
#           + augmented columns in campaign_patches.csv + per_patch_indicators.csv

# Stage 5: street-level solar envelope (winter / annual / summer)
python scripts/run_street_solar.py --site <site> --n-jobs -1
# Outputs → outputs/<site>/morphometrics/svf/svf_streets_solar.gpkg

# Stage 6: UMEP cross-validation (optional, ~10–45 min per site)
python scripts/validate_svf_against_umep.py --site <site> \
  --pixel-size 1.0 --observer-height 1.5
# Outputs → outputs/<site>/morphometrics/svf/umep_validation/
```

12.4 Per-figure regeneration

Figure	Producer	Source path before TR copy
1 Study sites overview	python outputs/paper_figures/fig01_study_sites.py	outputs/paper_figures/exports/fig01_study_sites.png
2 Morphometric distributions	python outputs/paper_figures/fig02_morphometric_distributions.py	outputs/paper_figures/exports/fig02_morphometric_distributions.png
3 SVF- λ p coupling	python outputs/paper_figures/fig03_svf_lambda_coupling.py	outputs/paper_figures/exports/fig03_svf_lambda_coupling.png
4 Sampling design	python outputs/paper_figures/fig04_sampling_design.py	outputs/paper_figures/exports/fig04_sampling_design.png
5 Morphometric maps	python outputs/paper_figures/fig05_morphometric_maps.py	outputs/paper_figures/exports/fig05_morphometric_maps.png
6 SVF $\leftrightarrow$ solar dissociation	python outputs/paper_figures/fig06_terrain_aspect.py	outputs/paper_figures/exports/fig06_terrain_aspect.png
7 Vidigal solar envelope (winter / annual / summer)	python scripts/run_street_solar.py --site vidigal && python outputs/paper_figures/fig07_solar_envelope_vidigal.py	outputs/paper_figures/exports/fig07_solar_envelope_vidigal.png (+ fig07a/b/c_*.png standalone panels)
S — terrain-aspect spatial	python outputs/paper_figures/figS_terrain_aspect_spatial.py	outputs/paper_figures/exports/figS_terrain_aspect_spatial.png
S1 Correlation matrices	python outputs/paper_figures/figS1_correlation_matrices.py	outputs/paper_figures/exports/figS1_correlation_matrices.png
S2 Context extension	python outputs/paper_figures/figS2_context_extension.py	outputs/paper_figures/exports/figS2_context_extension.png
S3 Resolution sensitivity	python outputs/paper_figures/figS3_resolution_sensitivity.py	outputs/paper_figures/exports/figS3_resolution_sensitivity.png
S5 Wind roses	python outputs/paper_figures/figS5_wind_roses.py	outputs/paper_figures/exports/figS5_wind_roses.png
S6–S10 UMEP scatters (per site)	python scripts/validate_svf_against_umep.py --site <site>	outputs/<site>/morphometrics/svf/umep_validation/scatter.png
campaign-allocation summary	side-effect of scripts/run_campaign_sampling.py	outputs/comparative/final_allocation/fig_campaign_allocation_summary.png
cross-site feature space	side-effect of scripts/run_campaign_sampling.py	outputs/comparative/final_allocation/fig_campaign_cross_site_featurespace.png

After regenerating, copy the figure into `docs/technical_report/figures/` and rebuild the PDF:

```
cp outputs/paper_figures/exports/figXX_*.png docs/technical_report/figures/
python docs/technical_report/build_pdf.py
```

The pre-commit hook (`.claude/hooks/check_report_sync.py`) blocks commits that stage `technical_report.md` without `technical_report.pdf` , or vice versa.

Known orphans: four pilot-summary figures (`fig_all_sites_patches` , `fig_candidate_pool` , `fig_patch_metrics_comparison` , `fig_strata_heatmap`) survive on disk and are referenced from §6, but their producer was deleted in the April 2026 `src/patch_selection/` removal and has not been re-implemented. Regenerating them is in the next-steps backlog.

12.5 Per-table regeneration

Table	Source	One-liner
§1 Study Sites	<code>outputs/<site>/morphometrics/grid/grid_metrics.gpkg</code>	<code>geopandas.read_file(...)</code> then aggregate <code>H_mean</code> , <code>lambda_p</code> , etc.
§2.3 Wind table	<code>data/<site>/wind_rose.json</code>	<code>json.load(...)</code> → fields <code>n_records</code> , <code>calm_fraction</code> , <code>frequencies</code> , <code>year_range</code>
§4.2 / §10.3 UMEP cross-val	<code>outputs/<site>/morphometrics/svf/umep_validation/summary_stats.csv</code>	<code>pandas.read_csv(...)</code>
§5.2 Typology medians	<code>concat outputs/<site>/morphometrics/grid/grid_metrics.gpkg</code> across sites, group by typology (Vidigal+Rocinha = Hillside, CDA = Mixed, RdP+Maré = Flatland), <code>.median()</code>	
§6.4 Campaign allocation	<code>outputs/comparative/final_allocation/campaign_allocation_table.csv</code> , <code>campaign_strata_summary.csv</code>	
§6.5 Rectangular-domain audit	<code>outputs/comparative/cfd_methodology/audit_v1.csv</code> (one row per patch, fields <code>domain*_m</code> , <code>domain_blockage_ratio</code> , <code>domain_blockage_ok</code> , <code>source_data_required_m</code> , <code>source_data_extent_m</code> , <code>eligible</code>). Producer: <code>scripts/audit_rectangular_domain.py</code> .	
§9 Validation Summary	each row's "Evidence" column points to the source file	

12.6 PDF rebuild

```
python docs/technical_report/build_pdf.py
# → 15.5 MB PDF, ~10 s on standard hardware (pandoc → WeasyPrint)
```

The build is deterministic given a fixed `technical_report.md` .

13. Failure modes & observability

What breaks at each pipeline stage, what success looks like, and how the validators surface drift before it ships.

13.1 Per-stage success / failure signals

Stage	Producer	Success signal	Failure signal	Validator
Site onboarding	<code>data/<site>/ per data/README.md</code> contract	All required files present; CRS = EPSG:31983; <code>wind_rose.json</code> <code>quality_flag = "measured"</code>	Missing files; mixed CRS; placeholder wind rose	<code>data-co</code> checker
Building extension	<code>scripts/build_extended_context.py --buffer 700</code>	<code>data/<site>/</code> <code>buildings_extended_700m.gpkg</code> and matching DTM written; building count > site-only count	Empty geometry; CRS mismatch; sentinel pixels (−9999) survive into the merged DTM	<code>data-co</code> checker
Rectangular-domain audit	<code>scripts/audit_rectangular_domain.py</code> then <code>scripts/migrate_indicators_rectangular_v1.py --apply</code>	<code>outputs/comparative/cfd_methodology/audit_v1.csv</code> written with 119 rows, all <code>eligible = true</code> ; both <code>campaign_patches.csv</code> and <code>per_patch_indicators.csv</code> carry the new <code>domain*_m</code> columns	Any <code>eligible = false</code> row; mismatch between audit row count and campaign size; <code>blocken_radius_required</code> column survives migration	n/a — <code>audit_v</code> per-strat
Morphometric grid	<code>scripts/run_morphometric_audit.py</code>	<code>grid_metrics.gpkg</code> written with all 20+ columns; <code>svf_count > 0</code> for every eligible cell; per-site PDF report rendered	NaN-heavy SVF column (passageway sampler failed); cells with <code>svf_count = 0</code> returning NaN; <code>report_render</code> fails on missing inputs	n/a — re site PDF
Pilot sampling	<code>scripts/run_pilot_sampling.py</code>	12–15 patches per site; ≥ 1 patch per non-empty stratum; min spacing ≥ 80 m	Stratum coverage gap; min spacing < 80 m; eligibility filter rejects > 95 % of cells	sampling agent
Campaign sampling	<code>scripts/run_campaign_sampling.py</code>	22–25 patches per site; SVF-priority weighting reflected in stratum totals; <code>eligible = true</code> in every row of <code>audit_v1.csv</code> after the rectangular-domain audit	Spacing collision; stratum over- or under-allocation; <code>eligible = false</code> rows in audit output	sampling agent
Wind rose ingestion	<code>scripts/build_wind_rose.py</code>	<code>wind_rose.json</code> with <code>quality_flag: "measured"</code> , <code>n_records ≥ 60,000</code> , full 8-direction frequency vector	<code>quality_flag</code> "placeholder-prior" (climatological prior never replaced); INMET date-format break post-2019 (silent zero rows)	wind-in agent
CFD ingestion	(CFD repo at <code>~/Airflow</code> writes; <code>src/cfd_integration/</code> reads)	All 8 wind directions present per patch; <code>sample_points.csv</code> rows ≥ 10 k; <code>summary.json</code> valid	Missing direction; off-axis directory name (e.g. <code>wind_017/</code>); column drift in CSV; $ U_{mag} - \sqrt{(U^2+V^2+W^2)} > 0.01$	<code>cfd-res</code> ingestor (auto-de native C Airflow-r layouts)
Annualised aggregation	<code>scripts/analyze_cfd_results.py</code>	Per-site <code>outputs/<site>/cfd_analysis/per_patch_indicators.csv</code> , <code>grid_with_cfd.gpkg</code> ; covered cells in 350–410 range per site (synthetic baseline)	Coverage anomaly (cells well outside 350–410); regression sign flips; weighting falls back to uniform when wind rose missing	n/a — v synthetic
Report drift	<code>docs/technical_report/technical_report.md</code>	Numerical claims trace to source; cross-references resolve; PDF rebuilt	Prose drift from data (the §6.5-class bug); <code>.md ↔ .pdf</code> desync; figure copy missing	report- auditor (time) + claims- (pre-rev)

13.2 The four validators

Read-only project subagents under `.claude/agents/` — invoke before shipping. Each returns a structured punch list; all four can be run in parallel.

- **data-contract-checker** — site-level contract per `data/README.md`. Flags missing files, CRS drift, wind-rose placeholder.
- **sampling-auditor** — 12-strata coverage, 80 m spacing, per-patch integrity, Blocken margin. Surfaces `docs/cfd_sampling_overrides.yaml` documented gaps as WARN, not FAIL.
- **report-sync-auditor** — pipeline / figure / sampling change in a diff that didn't update the technical report. Runs against `working`, `staged`, or any git ref range.
- **numerical-claims-auditor** — extracts every numerical claim from `technical_report.md` and verifies against source files. Targets the §6.5-class prose-drift bug. Run before sending the TR for external review.

13.3 Pre-commit hook

`.claude/hooks/check_report_sync.py`:

- **Blocks** a commit that stages `technical_report.md` without `technical_report.pdf` (or vice versa).
- **Warns** when a figure under `docs/technical_report/figures/` is staged without a `technical_report.md` change.
- **Warns** on `feat:` / `fix:` commits touching `src/` or `scripts/` without staging a corresponding `tests/` file.

The blocking rule is intentional: a stale PDF is strictly worse than no PDF, because external readers trust the rendered artefact.

13.4 Common failure patterns and where they were caught

Pattern	Caught by	Example
Prose drift (the canonical bug)	numerical-claims-auditor	The §6.5 Blocken miss (May 2026): constraint check correct, prose claim "≥ 150 m" wrong; actual minimum 114 m at RDP-P15
Producer drift in CFD output	cfid-results-ingestor	Synthetic generator's <code> U_mag </code> inconsistent with $\sqrt{U^2+V^2+W^2}$; fixed by deriving <code>U_mag</code> from perturbed components
Validation propagation miss	numerical-claims-auditor (Wave 2 sweep)	§6.5 was fixed; §9's row referring to the same Blocken margin still carried the old "150 m" — caught and propagated in commit 904040e
Stratum coverage gap	sampling-auditor	RdP SVF2_SLP2_LP2 : 7 eligible cells all within 80 m of pilot patches; downgraded from FAIL to WARN via <code>docs/cfid_sampling_overrides.yaml</code>
Module/package collision	pytest collection error	<code>src/exposure.py</code> shadowed by <code>src/exposure/</code> ; renamed to <code>src/exposure/sky_exposure.py</code>
INMET data quirks	wind-ingestion	Date format change post-2019 (YYYY-MM-DD → YYYY/MM/DD); accent-bearing column name <code>direção_horaria</code> ; server cuts > 5 GB transfers

Appendix A — Figure Index

#	Title	File
1	Study sites overview	<code>fig01_study_sites.png</code>
2	Morphometric distributions	<code>fig02_morphometric_distributions.png</code>
3	SVF– λ p structural coupling	<code>fig03_svf_lambda_coupling.png</code>
4	CFD sampling design	<code>fig04_sampling_design.png</code>
5	Morphometric spatial maps	<code>fig05_morphometric_maps.png</code>
S1	Correlation matrices (5 sites)	<code>figS1_correlation_matrices.png</code>
S2	Extended context validation	<code>figS2_context_extension.png</code>
S3	Resolution sensitivity (10m vs 20m)	<code>figS3_resolution_sensitivity.png</code>
S5	Wind roses (5 sites, measured)	<code>figS5_wind_roses.png</code>
S6	UMEP cross-validation of SVF (Vidigal)	<code>figS6_umep_validation.png</code>
S7	UMEP cross-validation of SVF (Maré)	<code>figS7_umep_validation_mare.png</code>
S8	UMEP cross-validation of SVF (Rocinha)	<code>figS8_umep_validation_rocinha.png</code>
S9	UMEP cross-validation of SVF (Rio das Pedras)	<code>figS9_umep_validation_riodaspedras.png</code>
S10	UMEP cross-validation of SVF (Complexo do Alemão)	<code>figS10_umep_validation_complexo_do_alemao.png</code>
—	Strata heatmap	<code>fig_strata_heatmap.png</code>
—	Candidate pool breakdown	<code>fig_candidate_pool.png</code>
—	Campaign allocation summary	<code>fig_campaign_allocation_summary.png</code>
—	Cross-site feature space	<code>fig_campaign_cross_site_featurespace.png</code>
—	All sites patch overview	<code>fig_all_sites_patches.png</code>
—	Patch metrics comparison	<code>fig_patch_metrics_comparison.png</code>

All figures in `docs/technical_report/figures/` .

Appendix B — Commit History (Relevant to CFD Campaign)

```
4dac8e8 feat: CFD integration module + Fig S3 resolution sensitivity
1890123 docs: update README and ROADMAP for v5.0.0 – CFD campaign complete
43f8554 feat: Nature Cities paper figures – 9 figures + shared style module
571ba47 feat: CFD sampling pipeline – stratified patch selection + campaign allocation
5e6e4c1 docs(figS3): canonicalize distribution-overlay as the main figure
9768e94 feat(wind-rose): scaffolding + template generator for CFD wind forcing
d477935 test(cfd-integration): 46 tests covering schema, IO, aggregation, metrics, weighting
```

Appendix C — Key References

- Blocken, B. (2015). Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations. *Building and Environment*, 91, 219–245.
 - COST Action 732 (2007). Best practice guideline for the CFD simulation of flows in the urban environment.
 - Stewart, I. D. & Oke, T. R. (2012). Local Climate Zones for Urban Temperature Studies. *Bulletin of the American Meteorological Society*, 93(12), 1879–1900.
 - Tregenza, P. R. (1987). Subdivision of the sky hemisphere for luminance measurements. *Lighting Research & Technology*, 19(1), 13–14.
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Appendix D — Engineering review checklist

For reviewers reading this document for the first time. The most useful feedback is **specific** and falls into one of the categories below; phrase observations against a section number and a concrete claim.

What kind of feedback is most valuable

Category	Examples of useful feedback
Methodology (highest priority)	Is the SVF–UMEP cross-validation in §10.3 a strong-enough benchmark, given the height-shift transform? Should the $\sigma_H \leftrightarrow H_{\text{mean}}$ correlation in §4.5 be reported pooled rather than per-site? Is the 12-stratum SVF $\times$ slope $\times$ λ_p grid the right axis set for <i>health-relevant</i> wind regimes, or is wind direction a missing axis?
Pipeline / reproducibility	Does §12 give you enough to actually reproduce a figure on a fresh clone? Are there missing dependencies, hidden setup steps, or unstated assumptions?
CFD contract	Read <code>src/cfd_integration/README.md</code> — is the input contract specified tightly enough that an independent OpenFOAM team could produce conforming output without back-and-forth? Are the 8-direction sample-point CSVs the right hand-off (vs full VTU)?
Sampling	Is 119 patches the right total for 5 sites? Are SVF-priority weights ($\times 2.0 / \times 1.0 / \times 0.8$) defensible? Is the 80 m maximin spacing a constraint or a target?
Numerical claims	Spot-checks against the source paths cited in §12.5 are welcome — the <code>numerical-claims-auditor</code> ran a sweep on 2026-05-03 and the §9-class propagation miss was the canonical bug. Other §6.5-class bugs may exist.
Code review	If something looks wrong on read, file it; the production-readiness pass focused on prose, not module-by-module code. The pre-commit <code>ruff check</code> is the only continuous lint gate.

What will not produce useful feedback at this stage

- Wordsmithing of methodology prose (the report goes to a journal manuscript next, where copy-editing happens).
- Suggestions to switch CFD code or turbulence model — the contract is in flight at MIT ORCD; changes require a full re-run cycle.
- Style preferences for figures — figure design is locked for the Nature Cities submission; only correctness matters here.

Reviewer's reading path ($\approx$ 1 working day)

Time	Read
10 min	This document's Exec Summary + §0 Glossary
30 min	§1 Study Sites · §2 Data Sources · §3 Data Preparation
60 min	§4 Morphometric Grid · §5 Cross-Site Morphology
90 min	§6 CFD Patch Sampling (the central methodological claim)
30 min	§7 CFD Integration Pipeline (read alongside <code>src/cfd_integration/README.md</code>)
30 min	§9 Validation Summary · §10 Known Limitations
30 min	§11 Next Steps · §12 Reproducibility · §13 Failure modes
60 min	Walk one site end-to-end against §12.3 — pick <code>vidigal</code> ; the smallest valid grid ($n = 1,503$ in §10.3) makes it the fastest
Closing	Cite specific §X.Y + claim when filing observations

How to file feedback

Severity	Channel
Methodological / numerical errors	GitHub issue with section reference + a one-line repro
Suggested next experiments	GitHub issue tagged <code>discussion</code>
Pipeline-contract questions	Email the author (<code>thermann.ai@gmail.com</code>) — answers usually need design context
Editorial / typographical	Direct message; bulk fixes are batched

Prepared in `docs/technical_report/`. Available in both markdown (`technical_report.md`) and PDF (`technical_report.pdf`) formats.

Regenerate the PDF:

```
python docs/technical_report/build_pdf.py
```

Pipeline: `pandoc` (markdown → HTML) then `weasyprint` (HTML → PDF). Requires `pandoc` and the `weasyprint` Python package.

Regenerate all figures:

```
for f in outputs/paper_figures/fig*.py; do python3 "$f"; done
```

Regenerate campaign allocation:

```
python scripts/run_campaign_sampling.py
```